

Hysteresis Indices and Beta of IBW sites: Curry, Silverado, and Lake Marguerite

2026-03-26

Aims to plot hysteresis per analyte per storm per site

Metrics

- *Hysteresis index* is calculated as the median (+/- 95% CI) across 2% intervals of discharge.
+HI = clockwise (source limitation or nearstream source)
-HI = counterclockwise (transport limitation or distal source)
- *Beta* is the slope of the CQ relationship on the rising limb of the storm
CIs on beta are bootstrapped 95% CI

NOTES

- Lake Marguerite does not have a lot of chems to work with. Do we know if there are more chems than the 624_runoff_chems.csv from the dropbox?
- Similarly, there is a gap of data in Silverado 2014-2018, was there less sampling in that time or is the data elsewhere?
- individual analyte outliers > 3sd above mean concentration per storm were removed for hysteresis and beta calculations, but remain for individual storm plotting at the bottom of the report
- Precipitation data was collected from <https://azmet.arizona.edu/azmet/27.htm> - precipitation gage at the Desert Ridge station

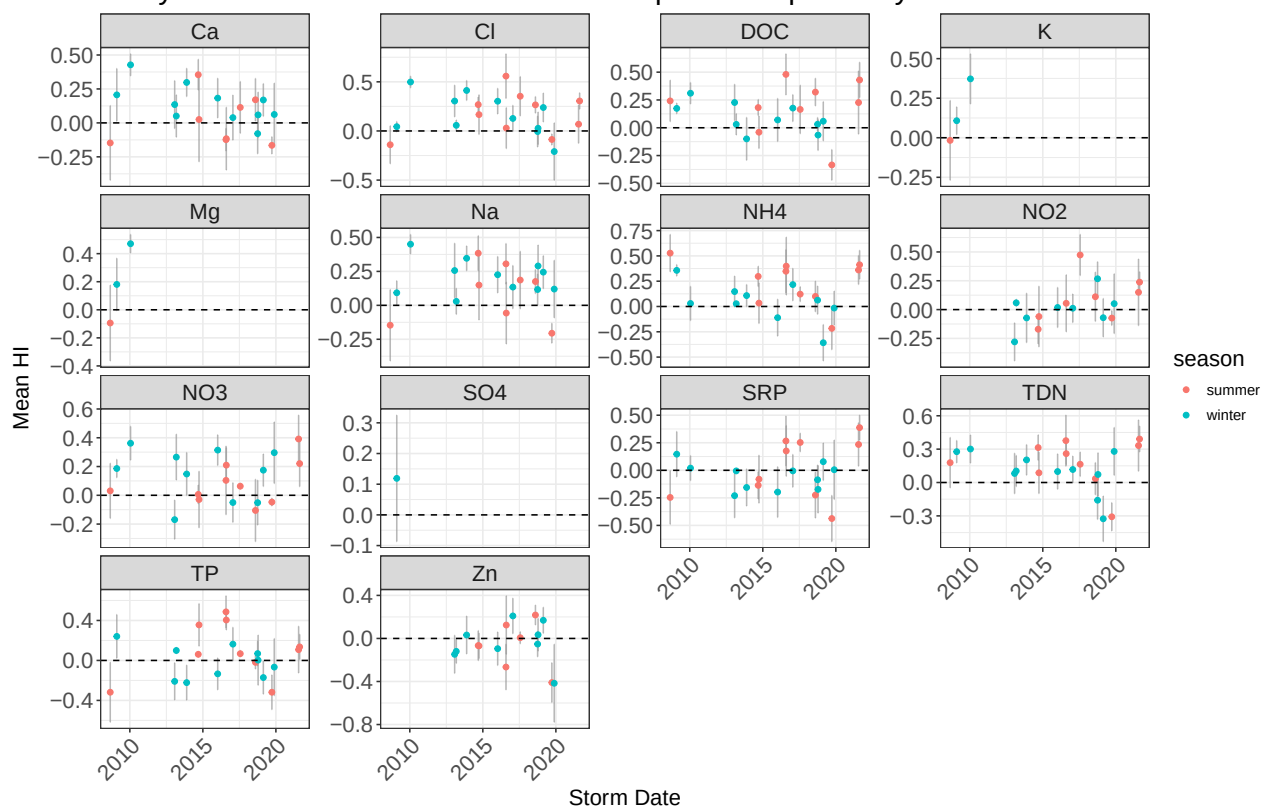
Data munging/cleaning

Workflow before this rmd, in order:

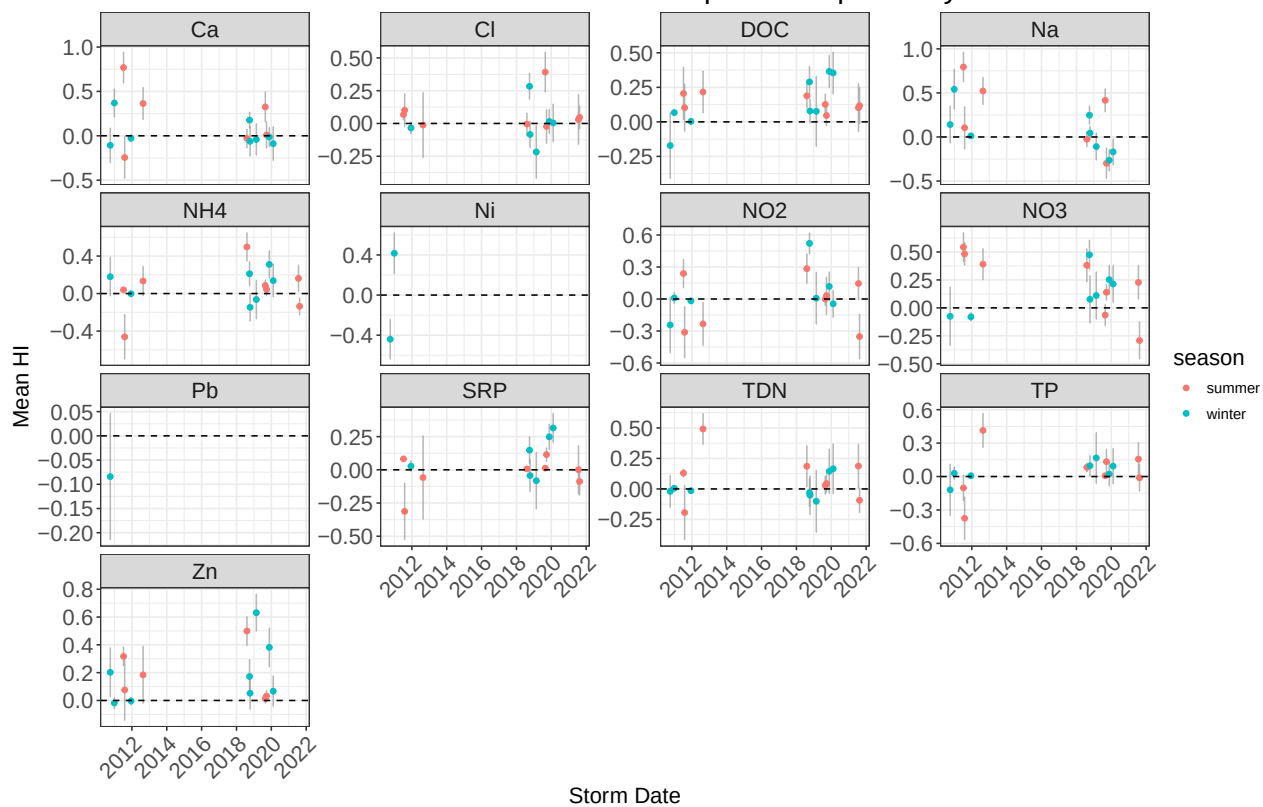
- q_storms.R: import discharge from Lake Margherite, Silverado, & Curry, delineate unique storms, and interpolate Q
- merge_Q_chems.R: merge discharge data with chem data per site, produce brief cq plots per analyte per site
- hysteresis_calc_2.R: calculate hysteresis based on python function, removed a lot of storms with not enough data. also calculates beta

Plotting Hysteresis

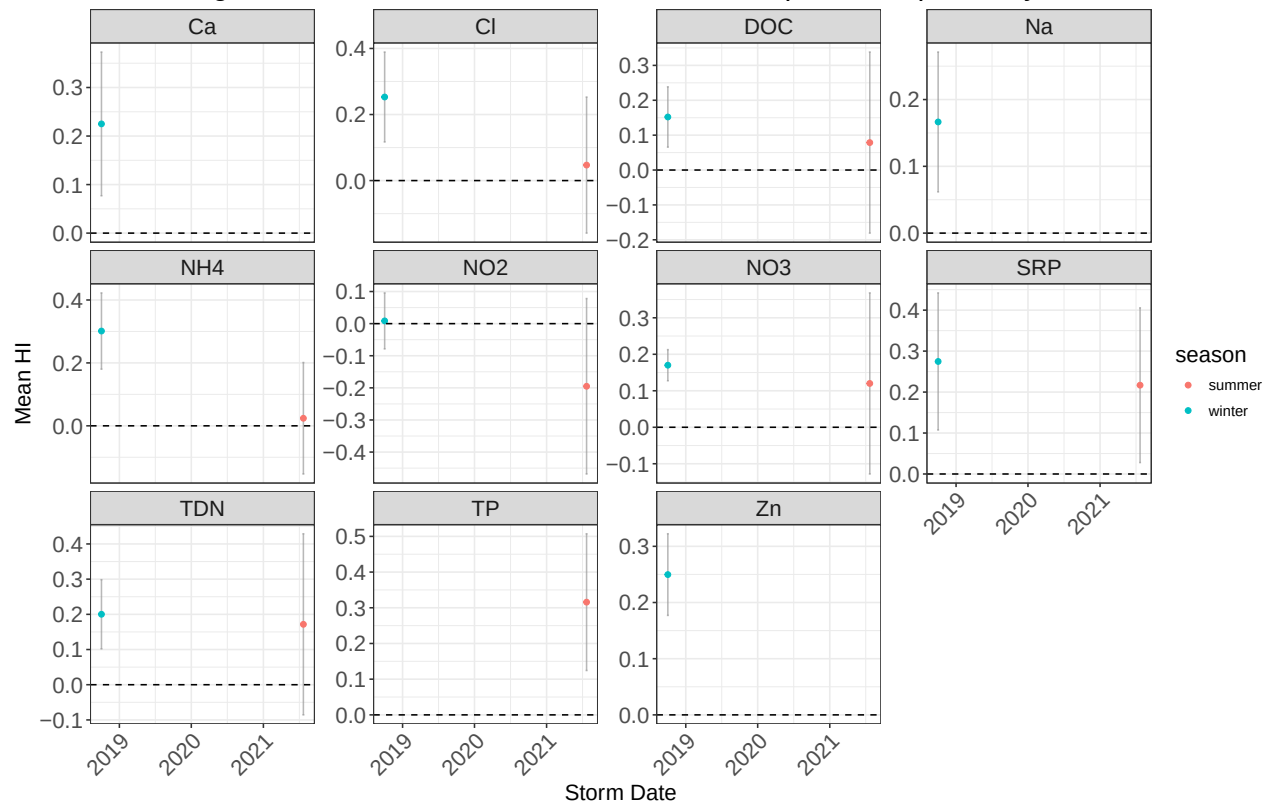
Curry – mean HI with confidence intervals per storm per analyte



Silverado – mean HI with confidence intervals per storm per analyte



Lake Marguerite – mean HI with confidence intervals per storm per analyte



Plotting Beta

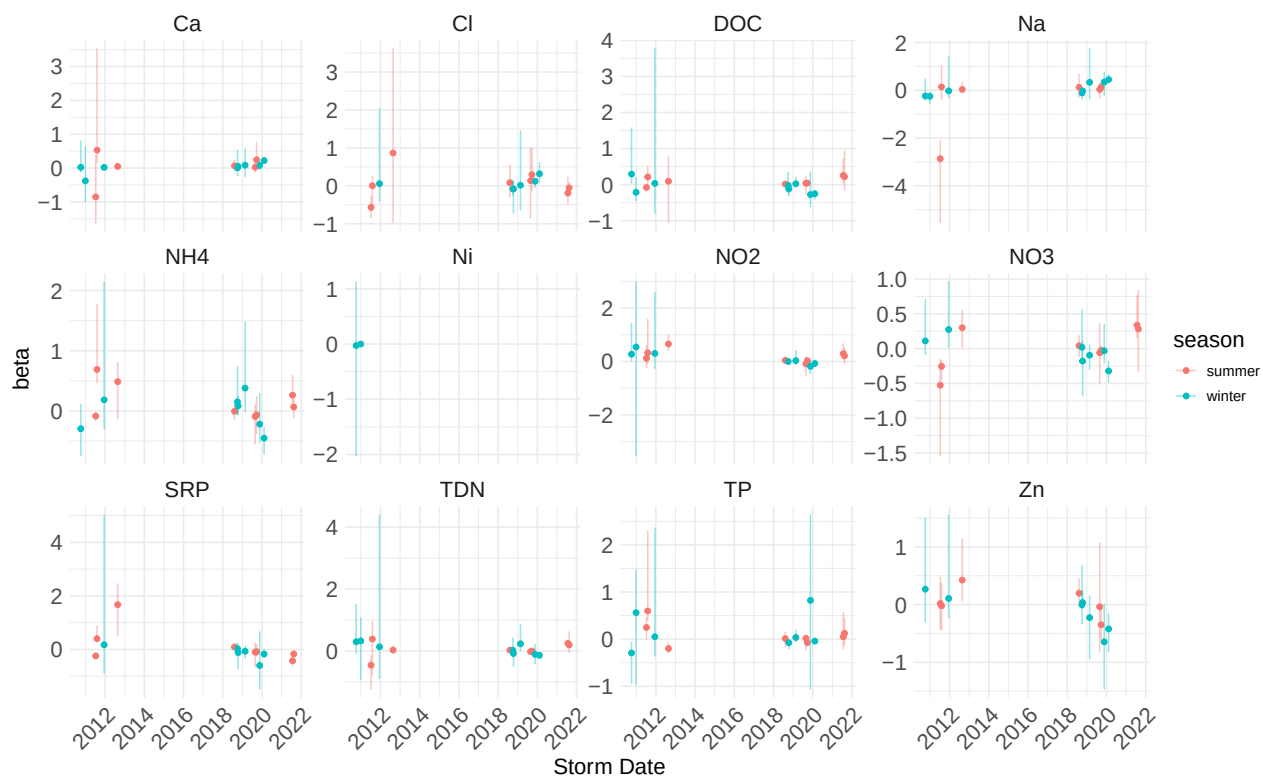
Curry Beta per storm per analyte

Bars are 95% confidence intervals



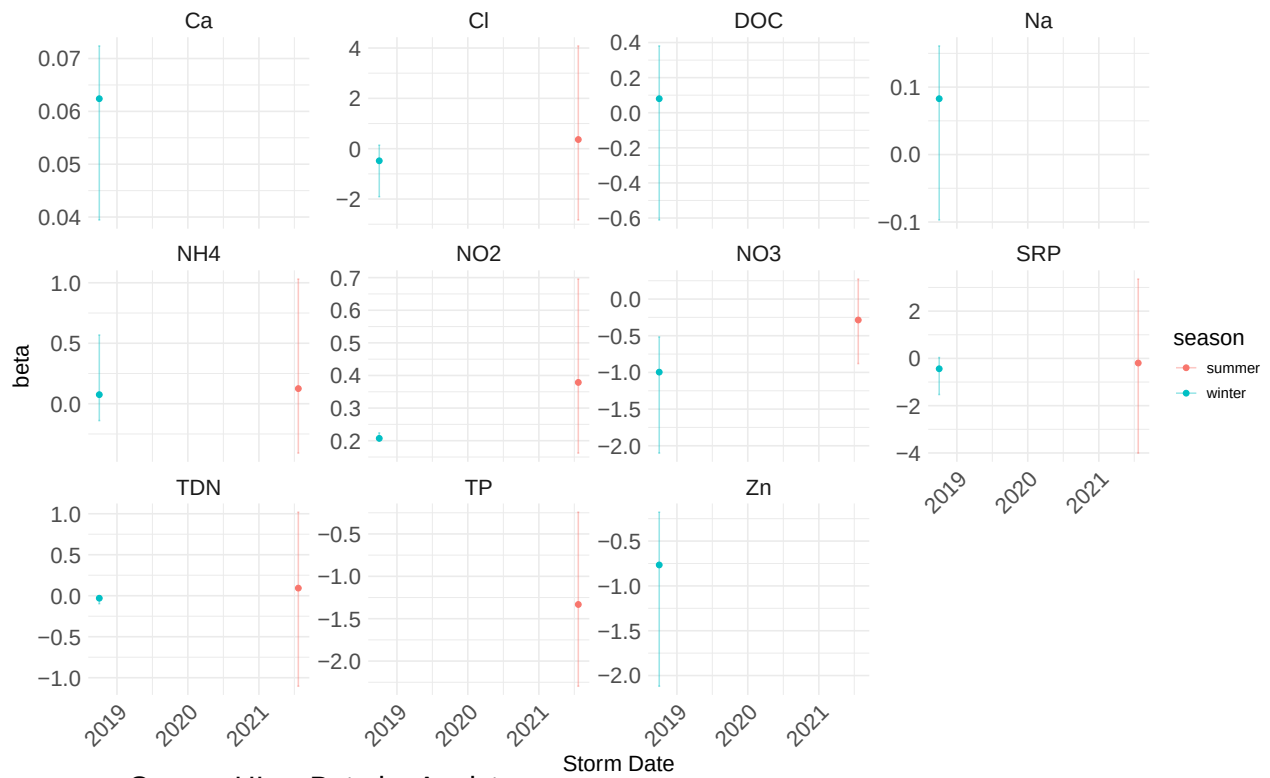
Silverado Beta per storm per analyte

Bars are 95% confidence intervals



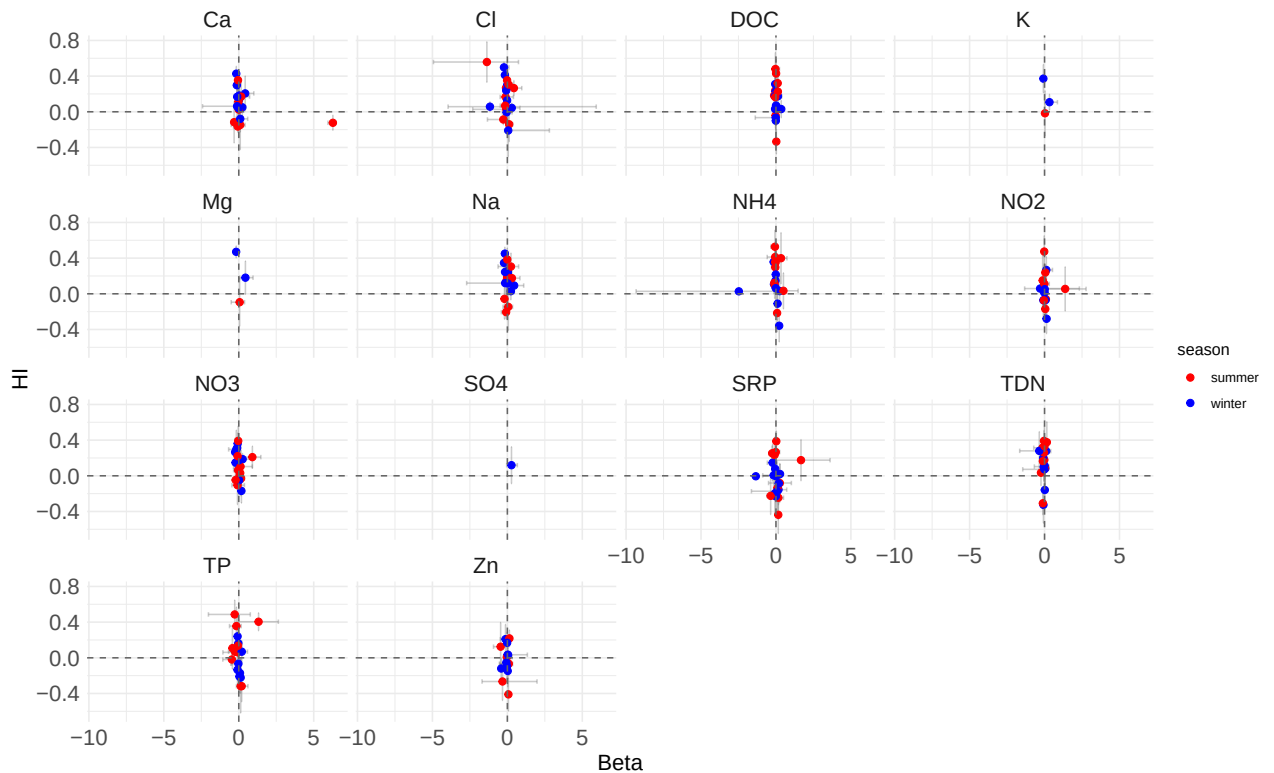
Lake Marguerite Beta per storm per analyte

Bars are 95% confidence intervals

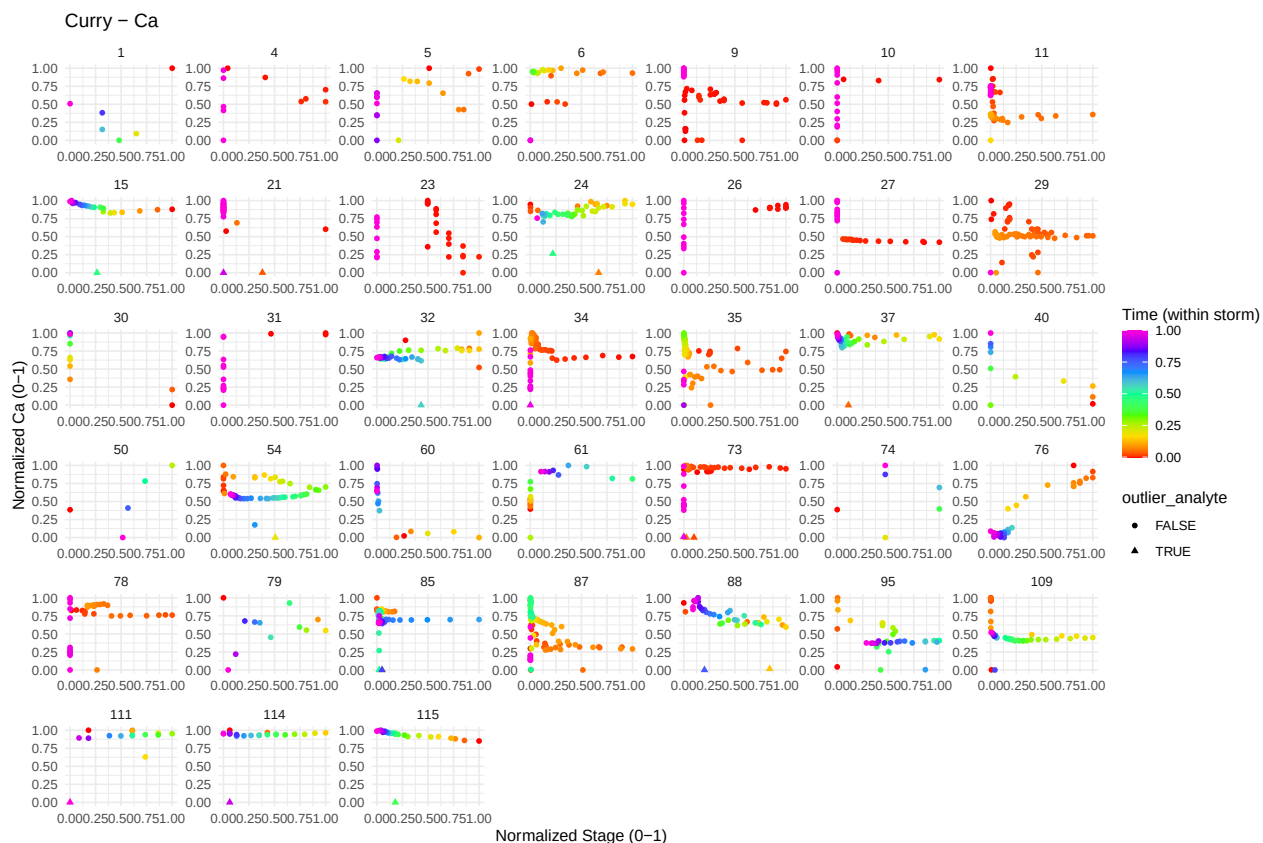


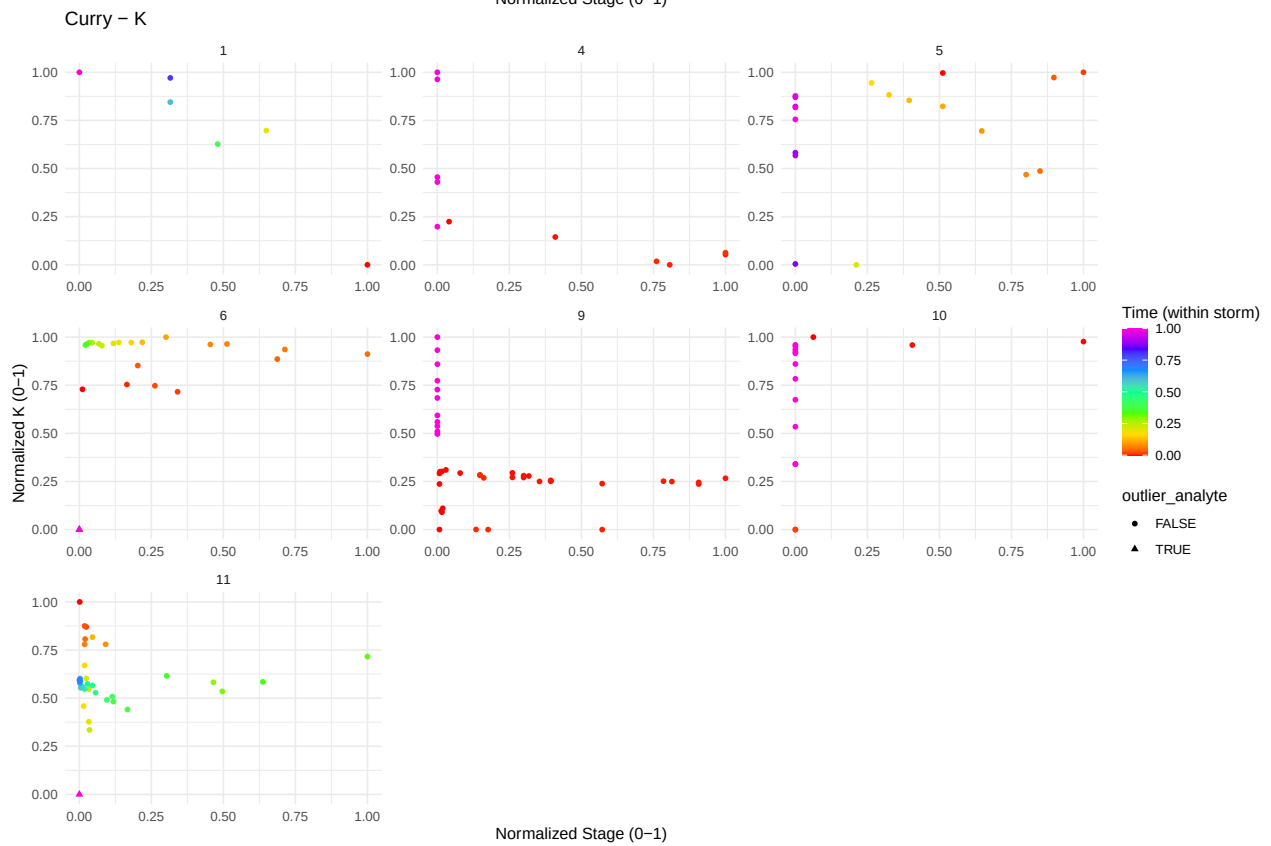
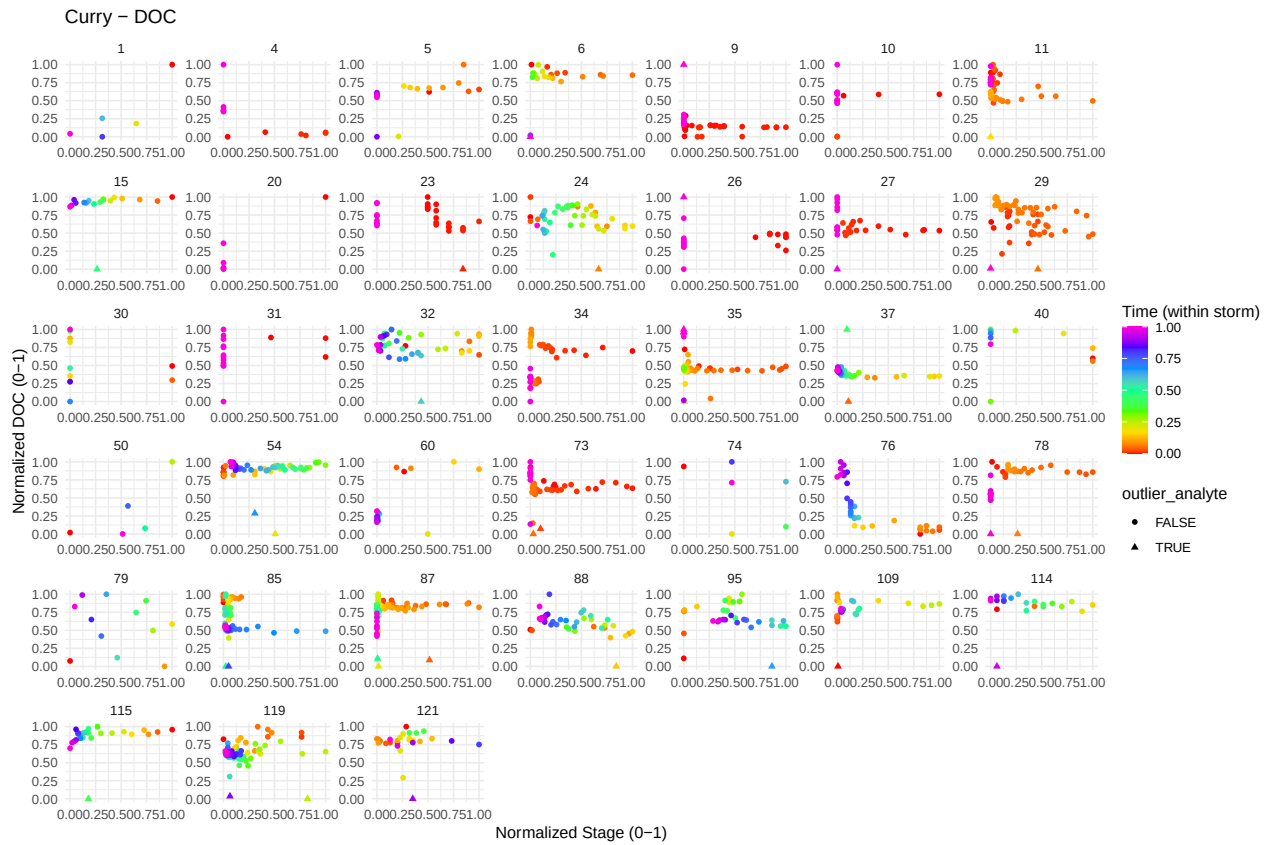
Curry – HI vs Beta by Analyte

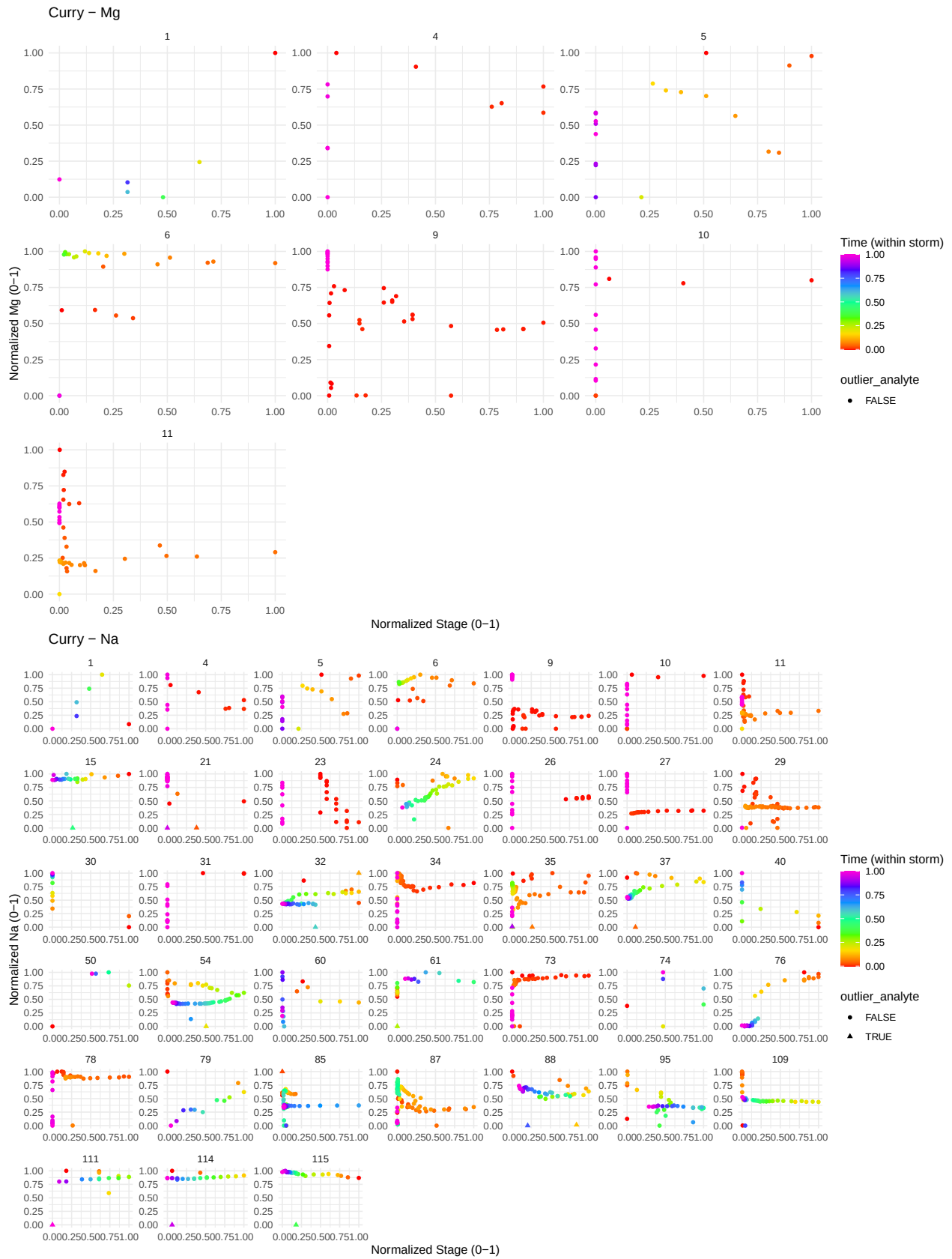
Each dot is an individual storm

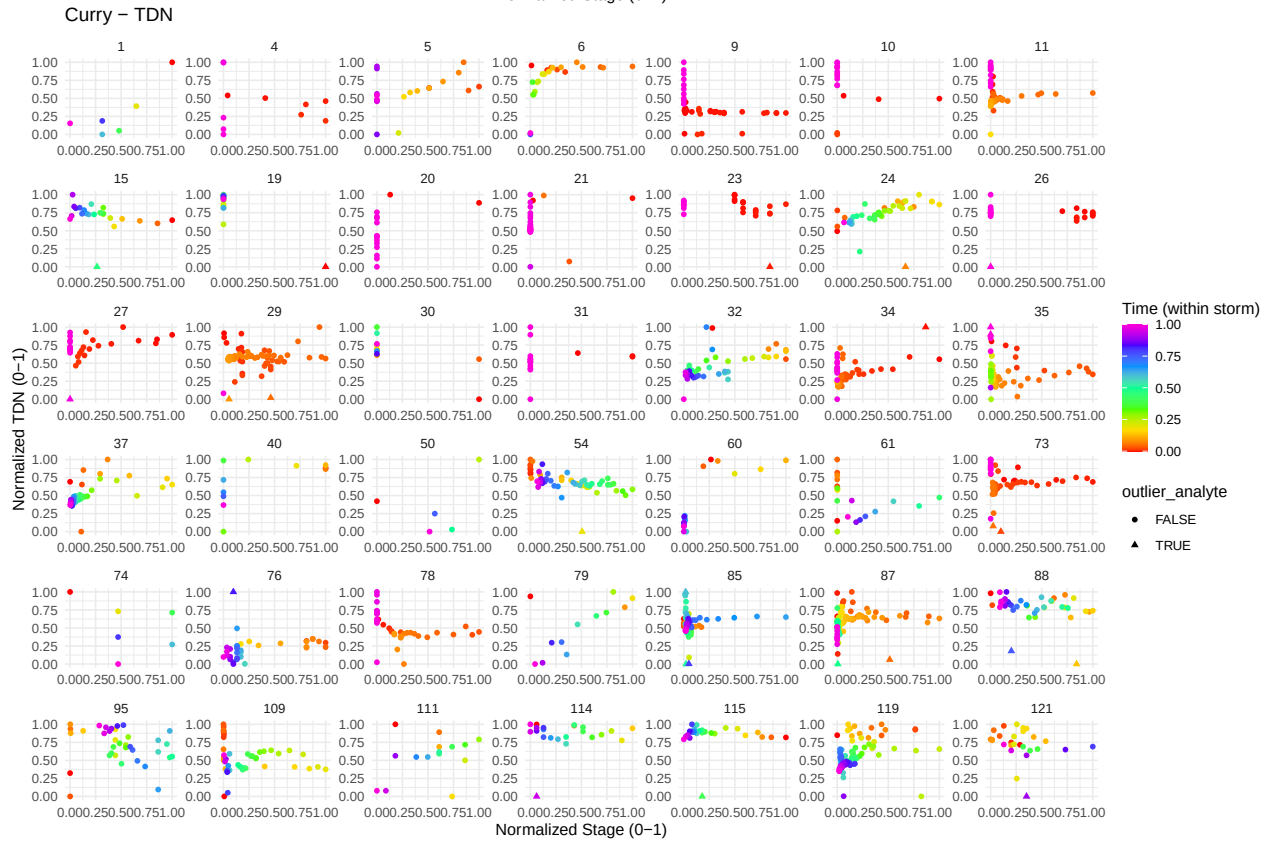
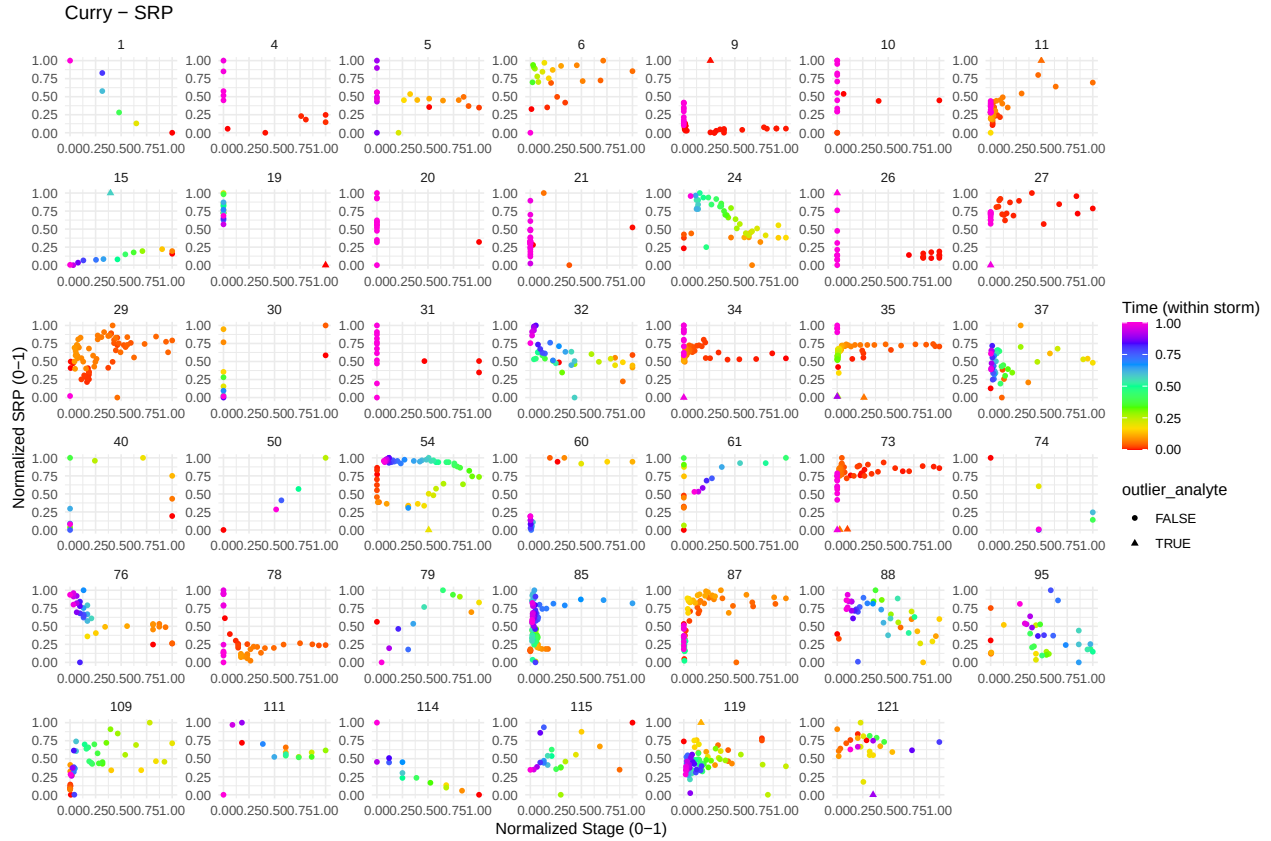


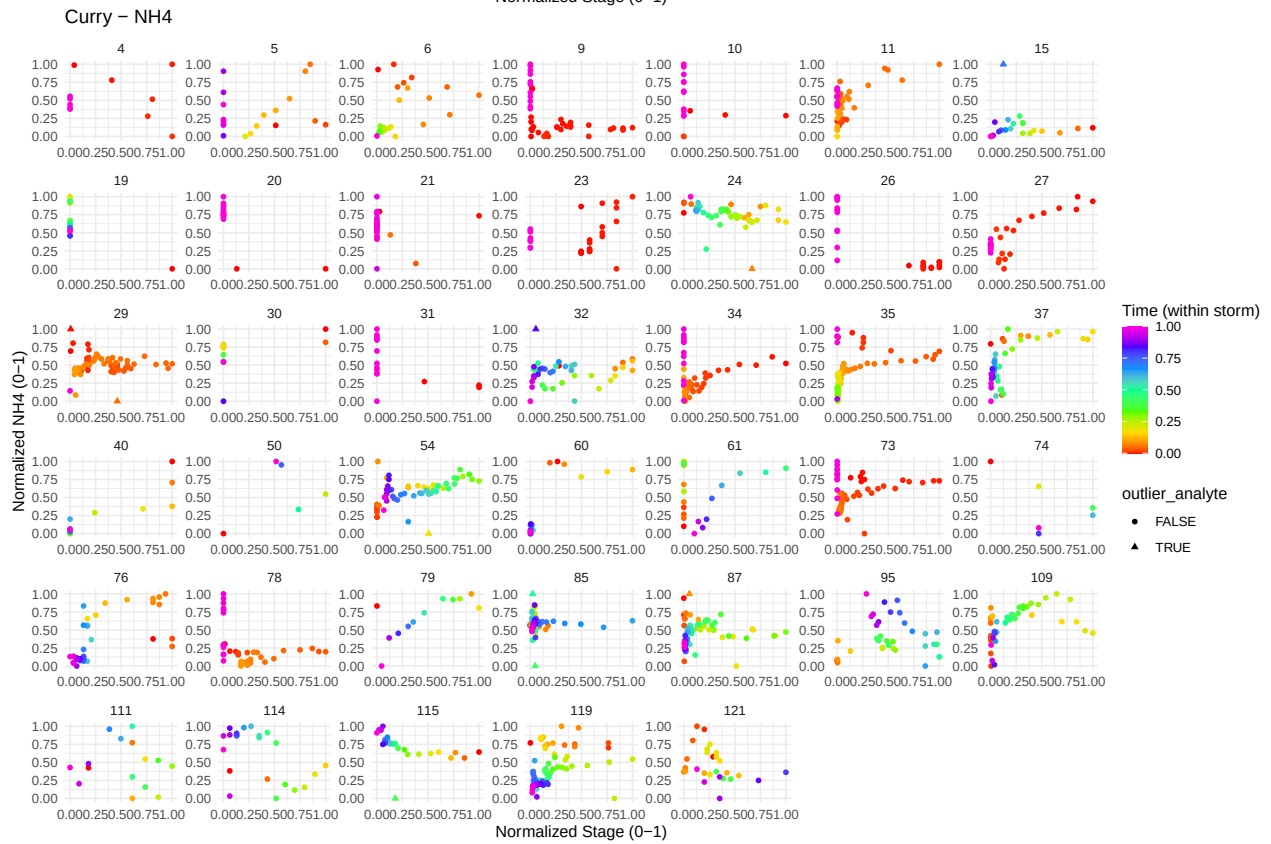
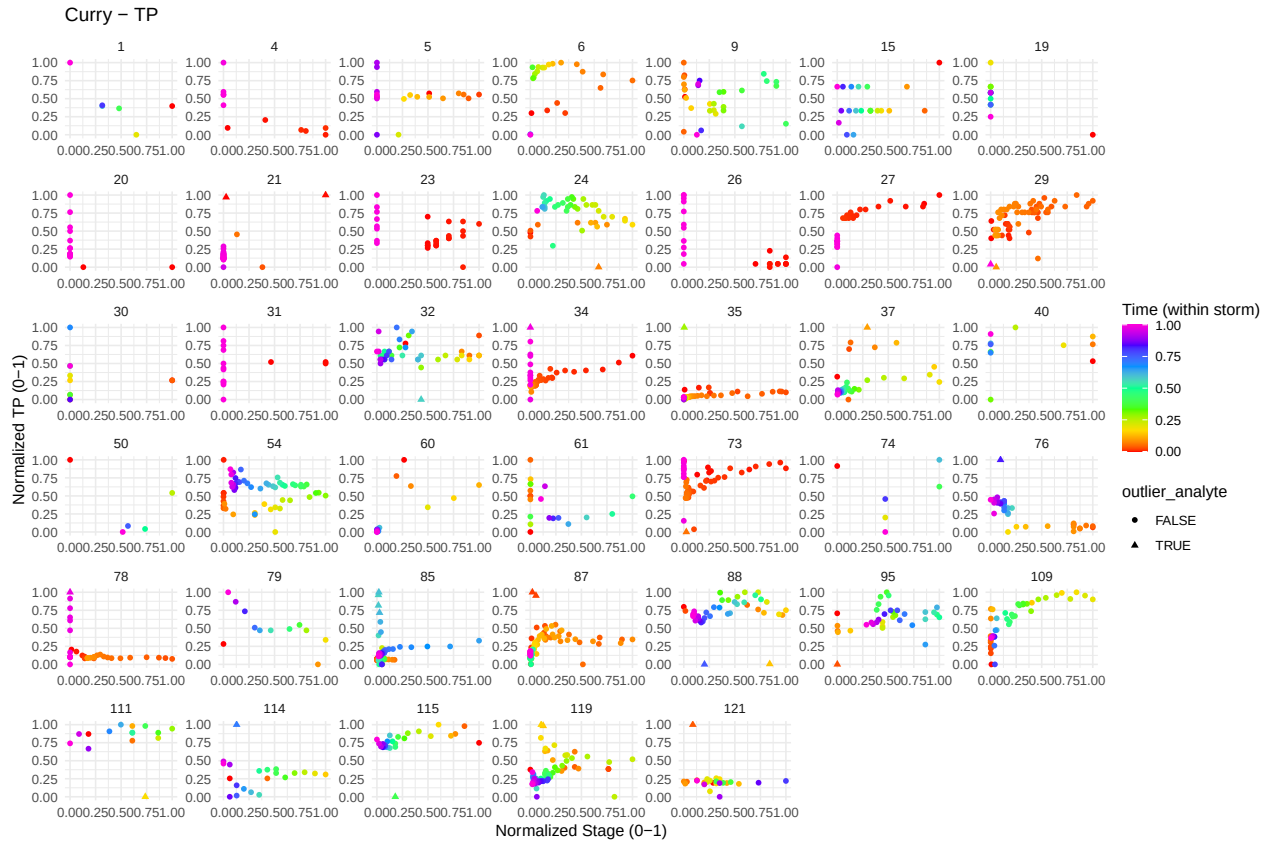
Plotting CQ

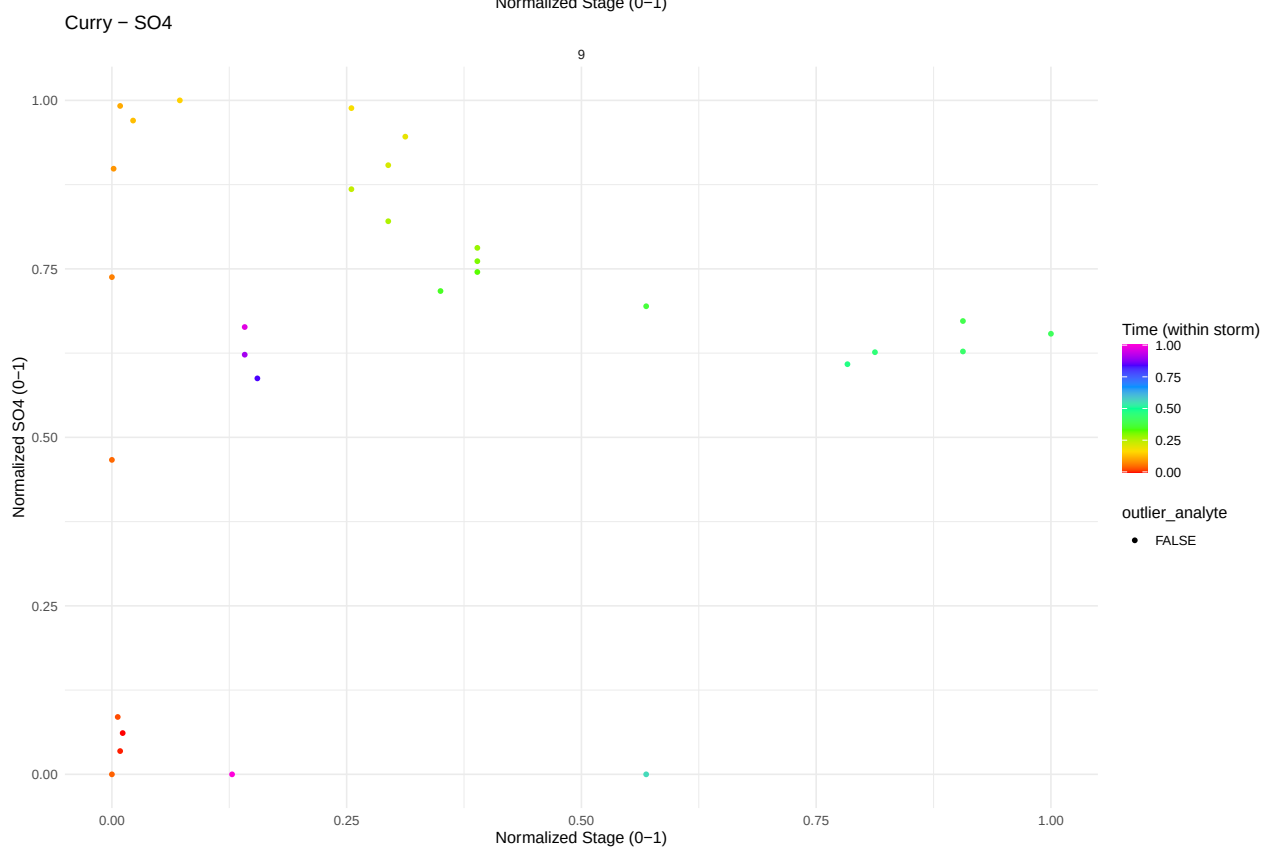
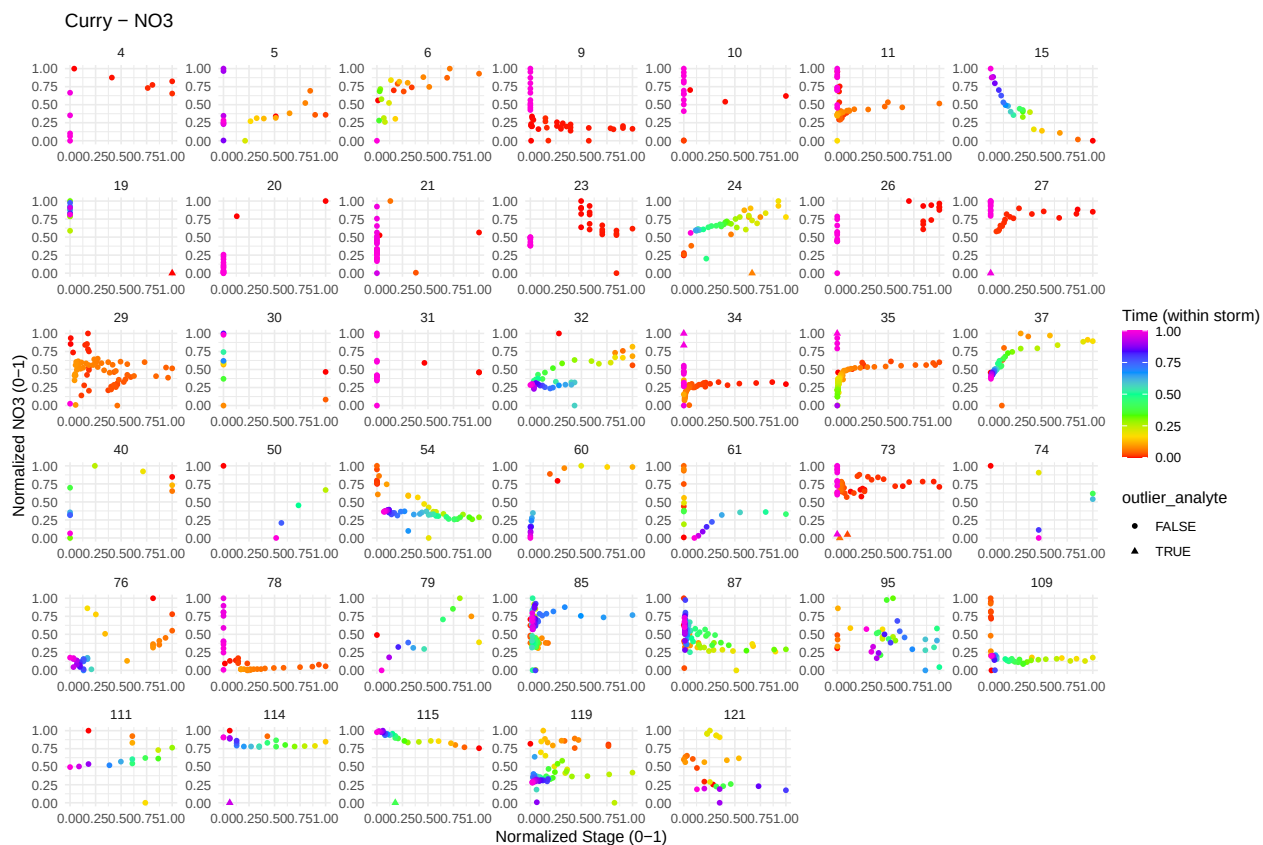


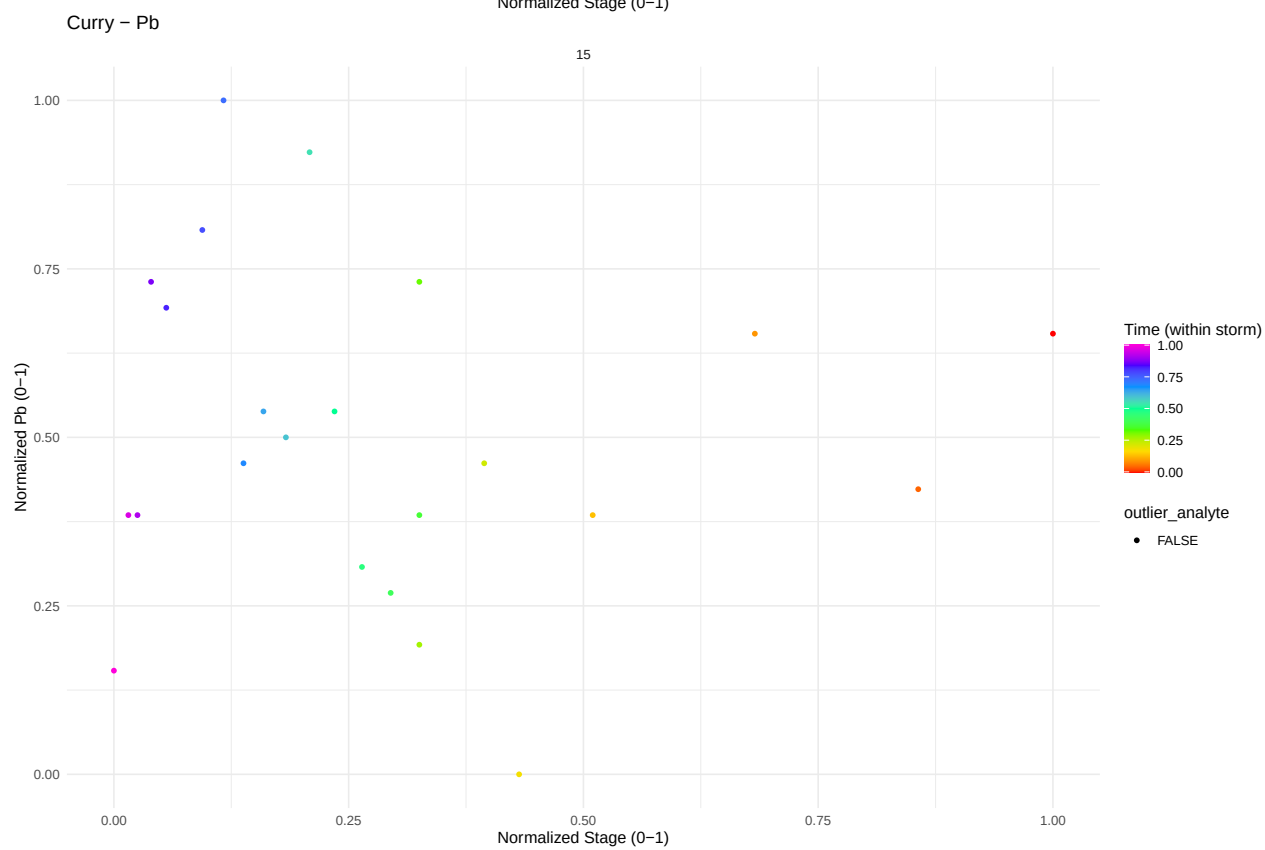
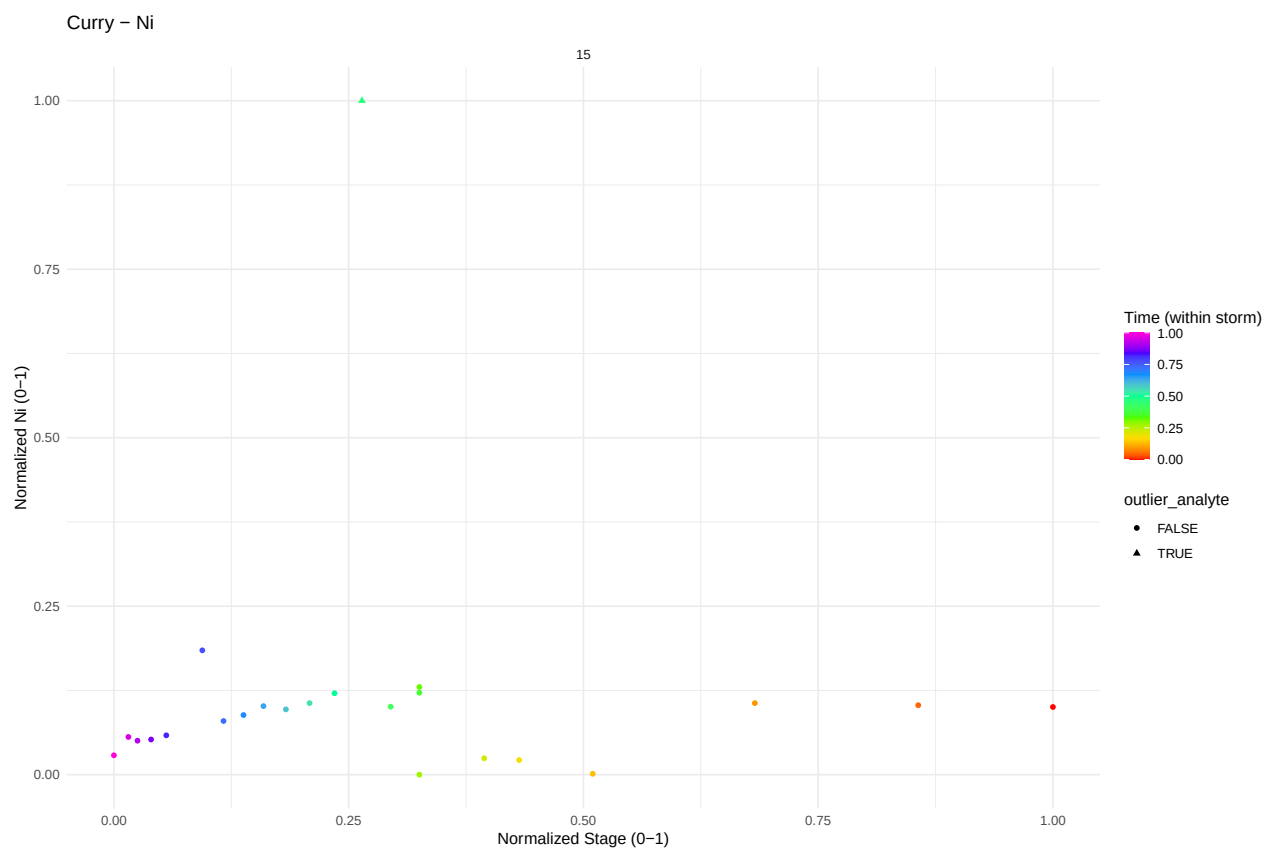


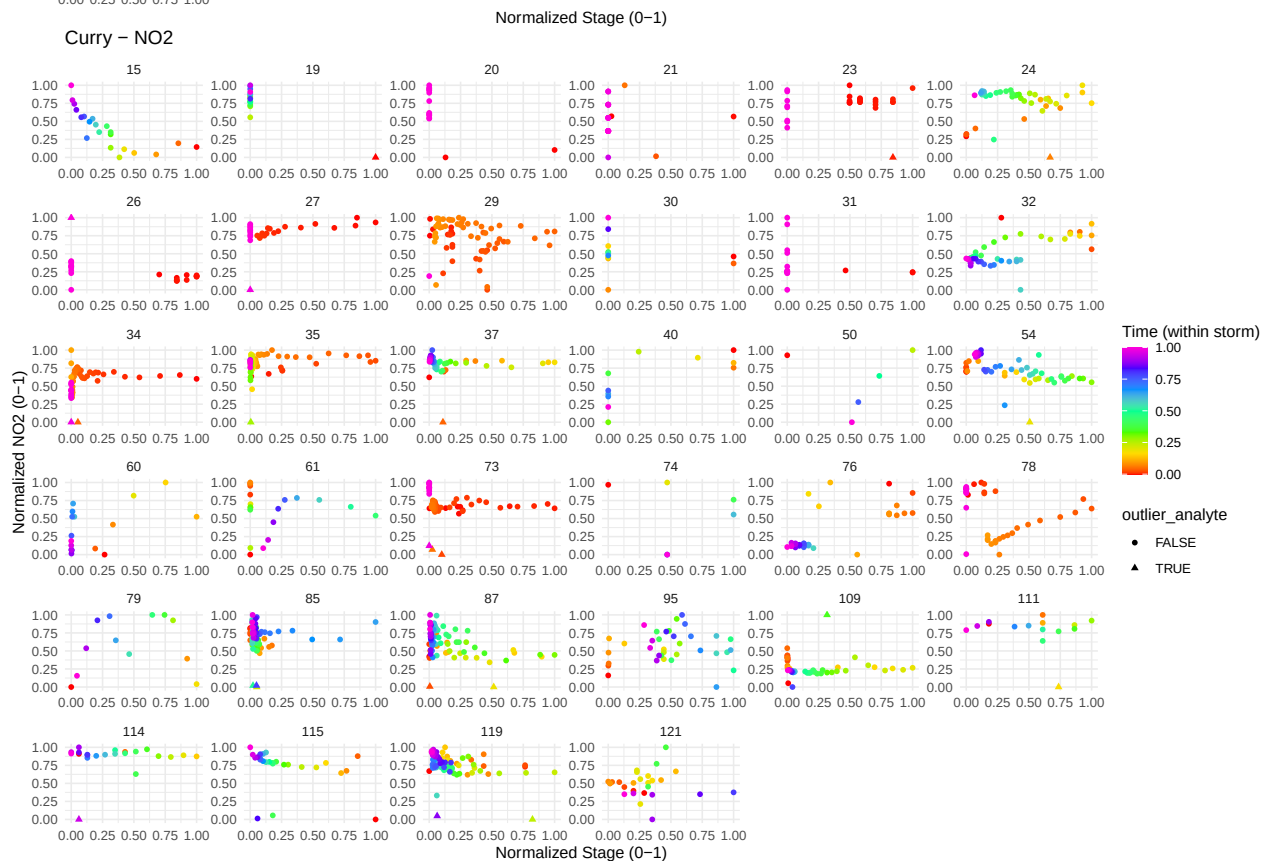
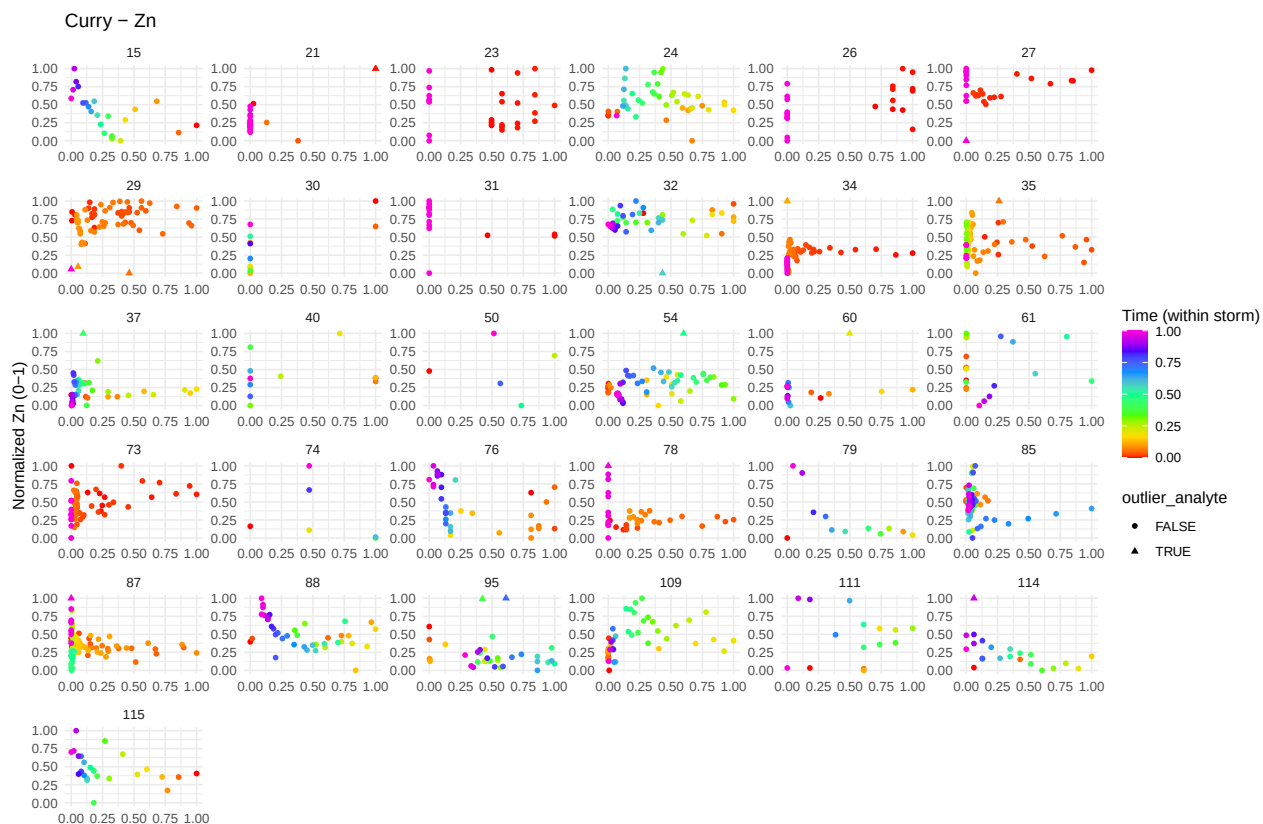


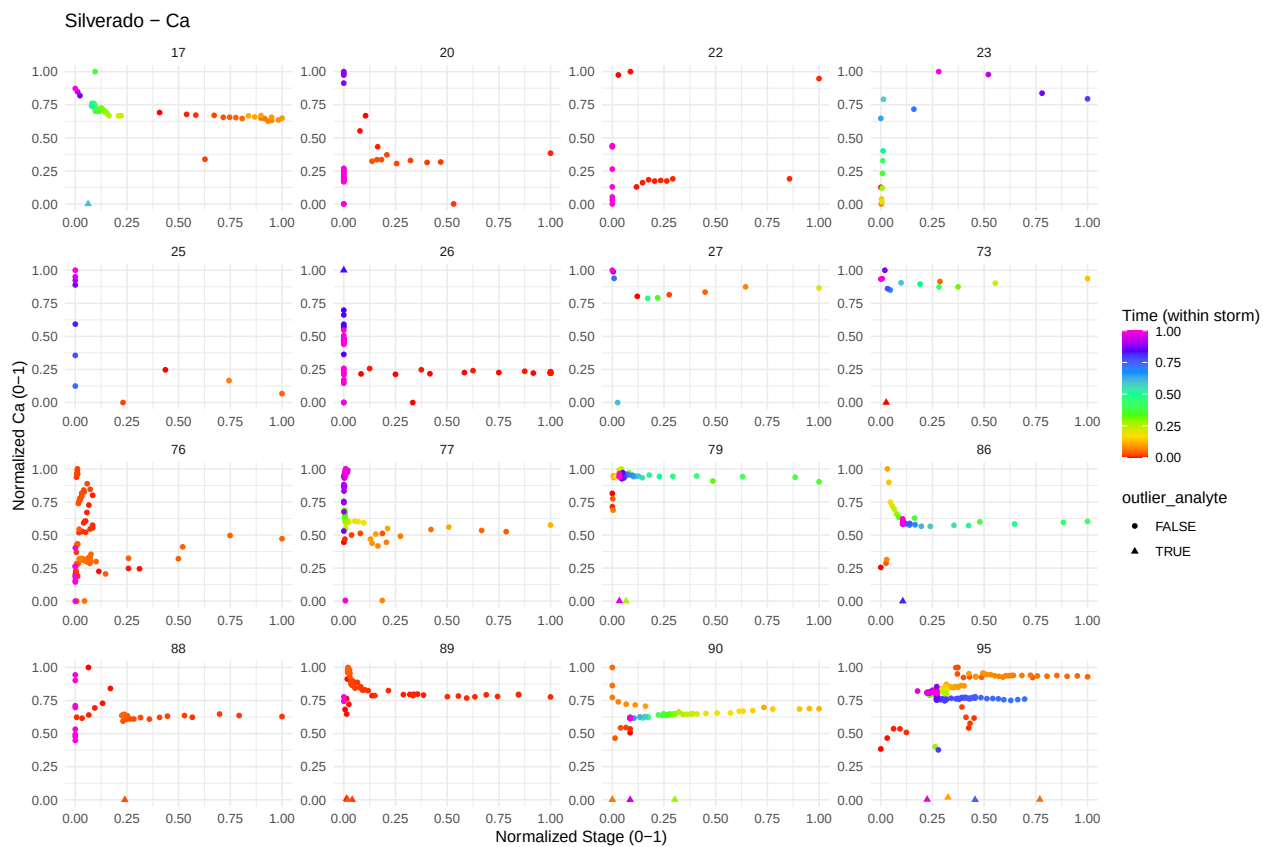




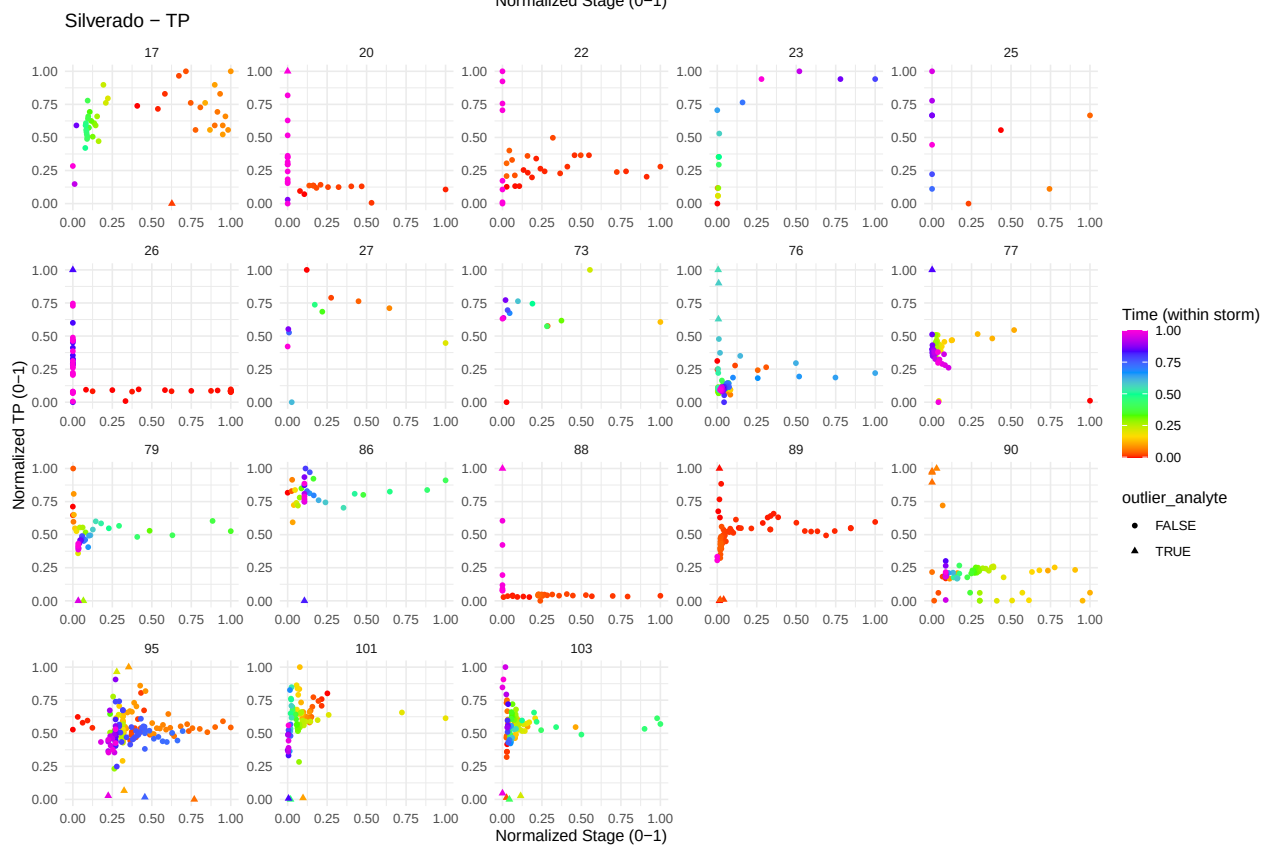
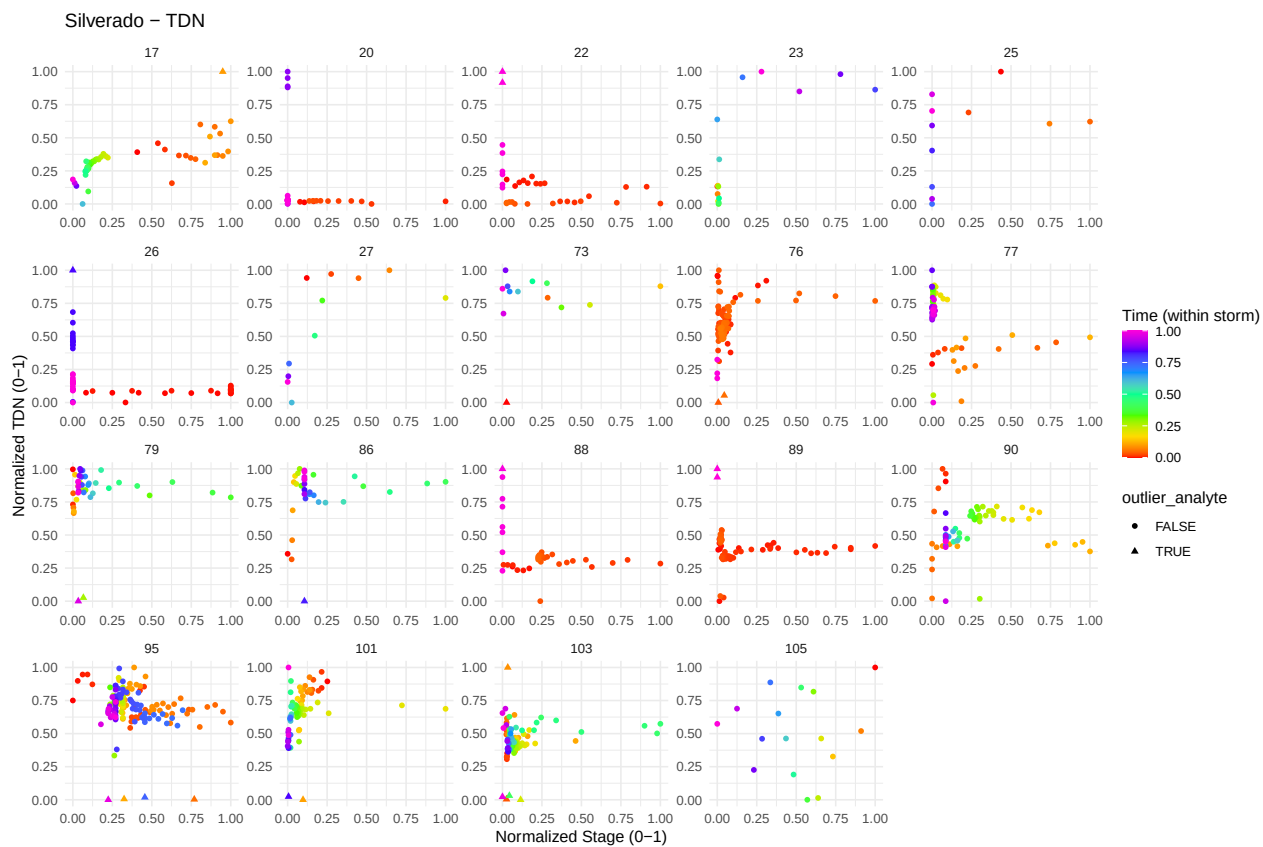


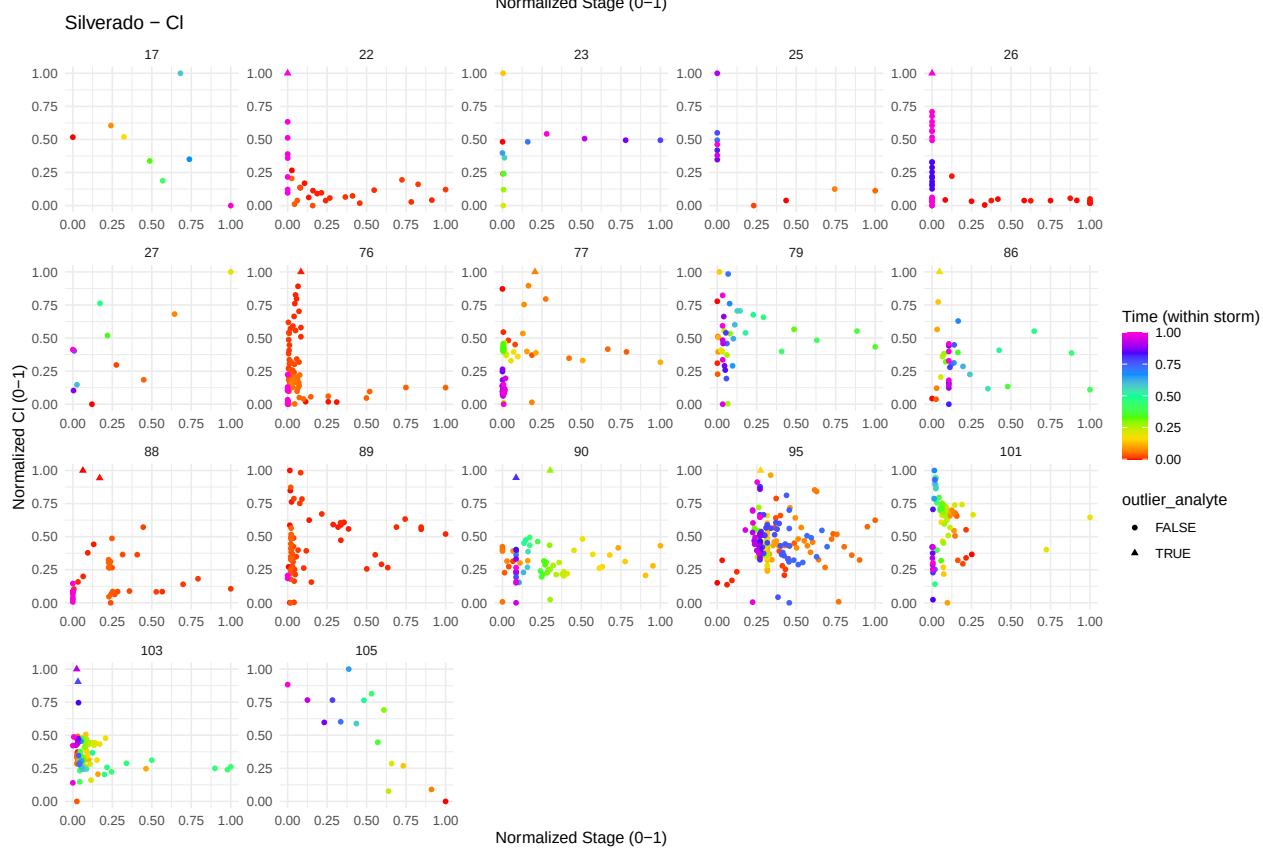
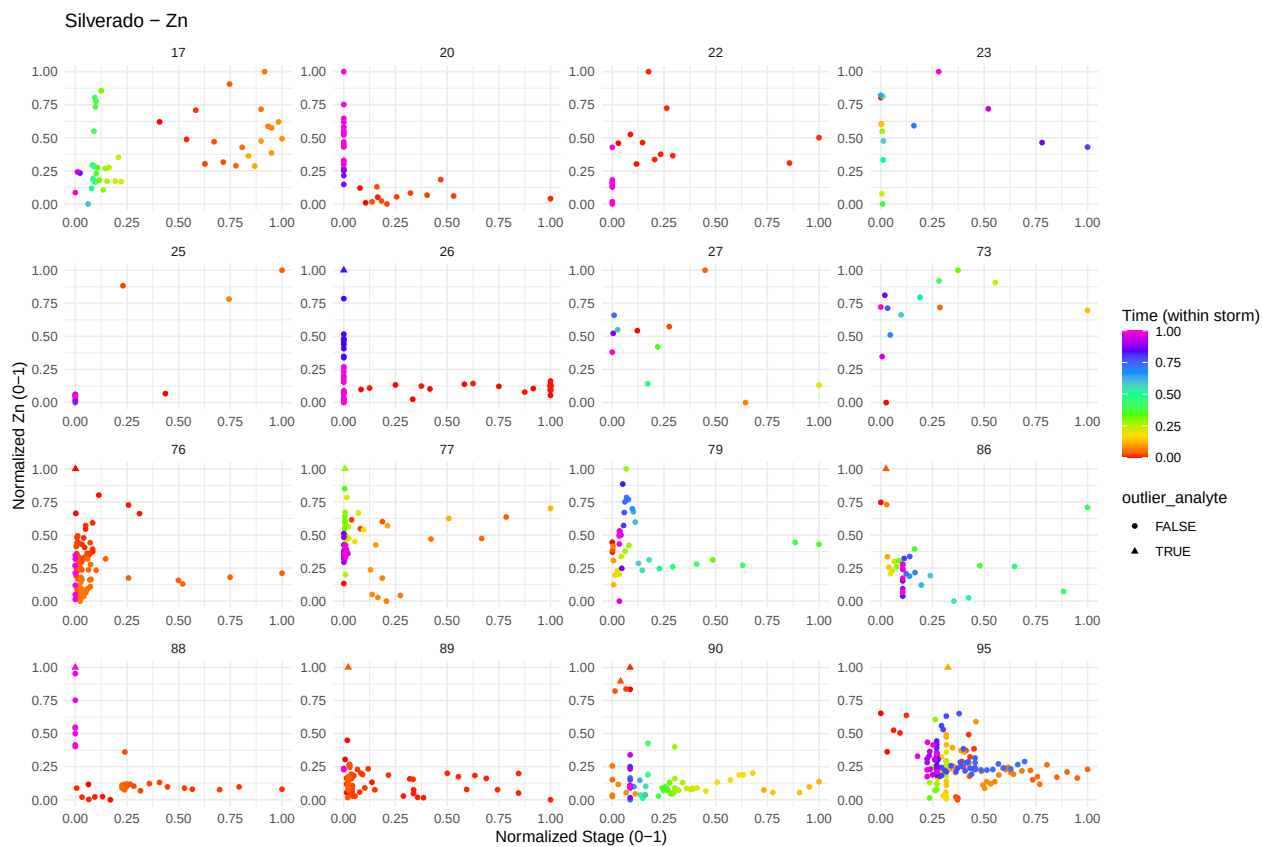


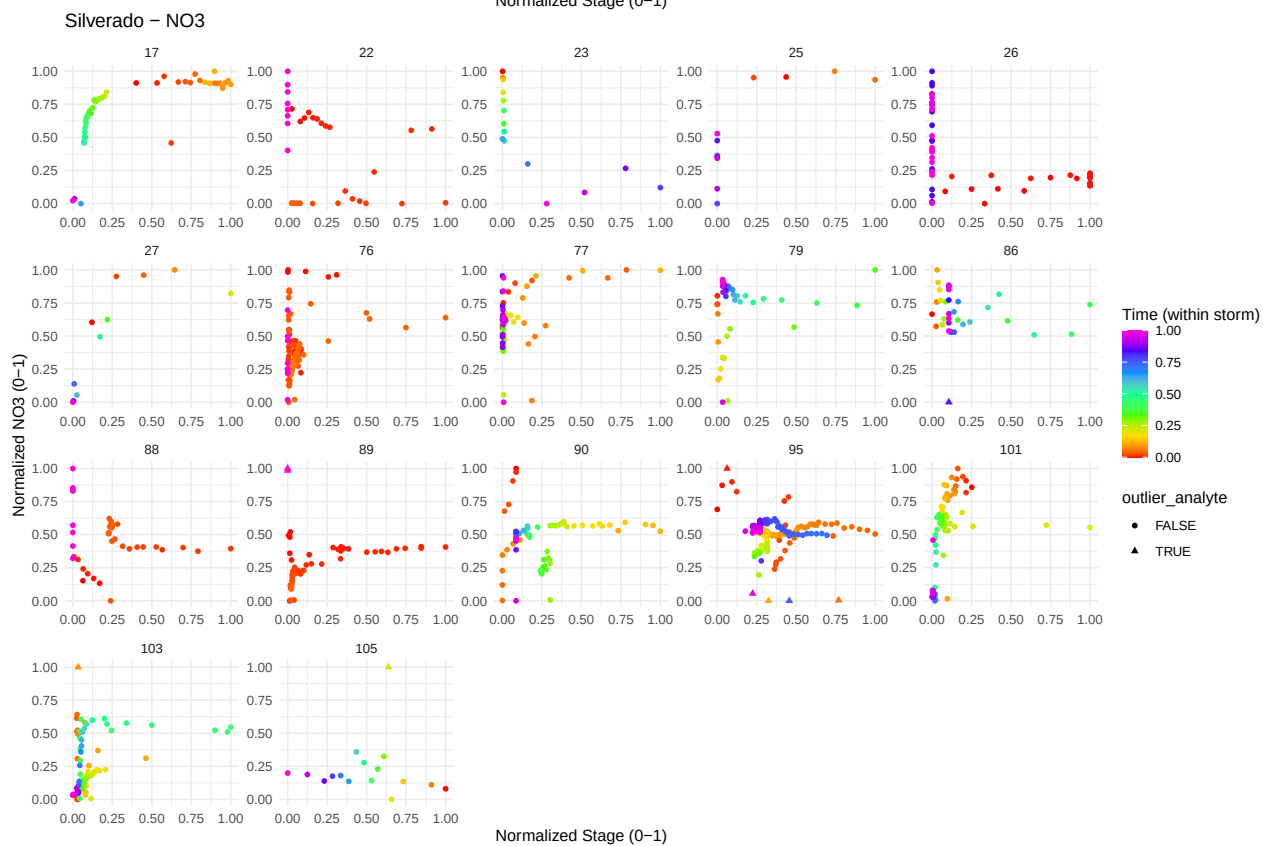


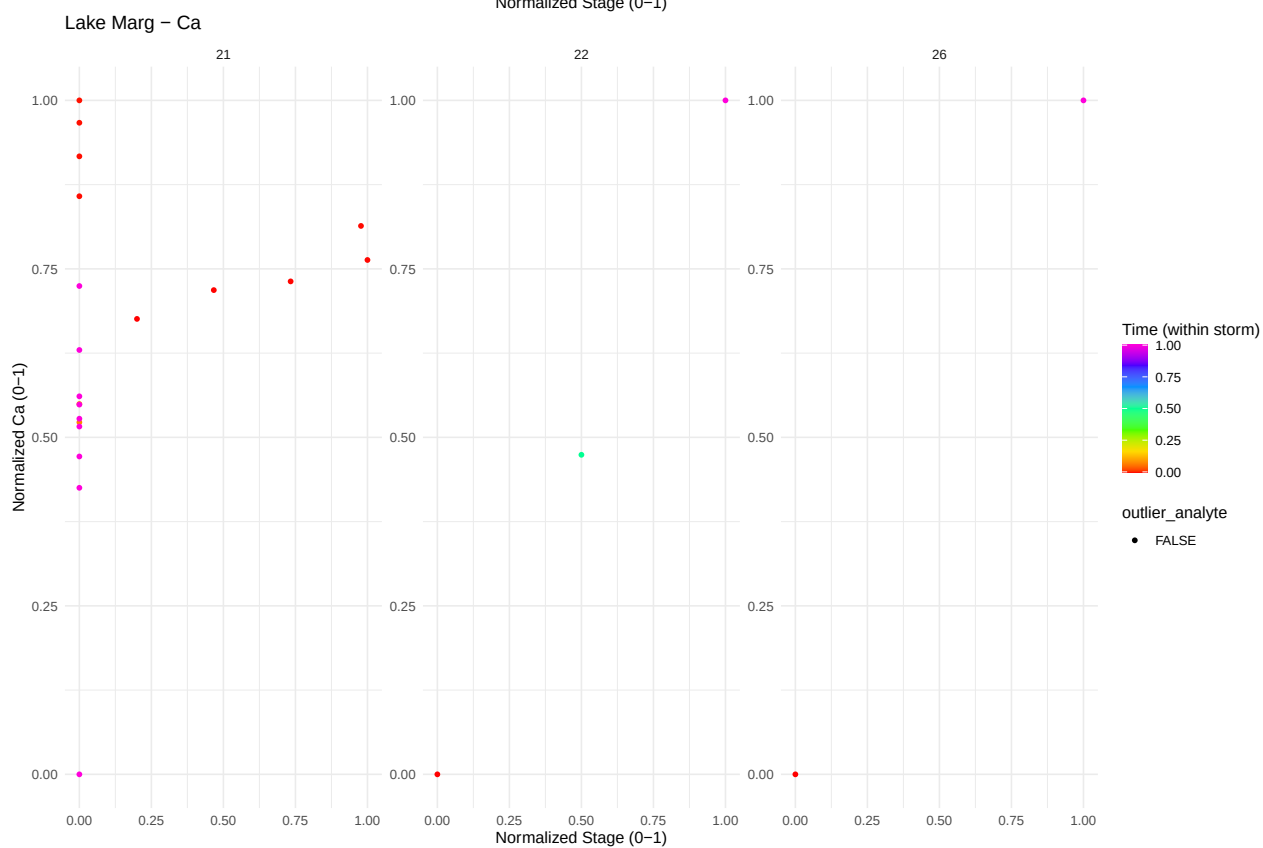
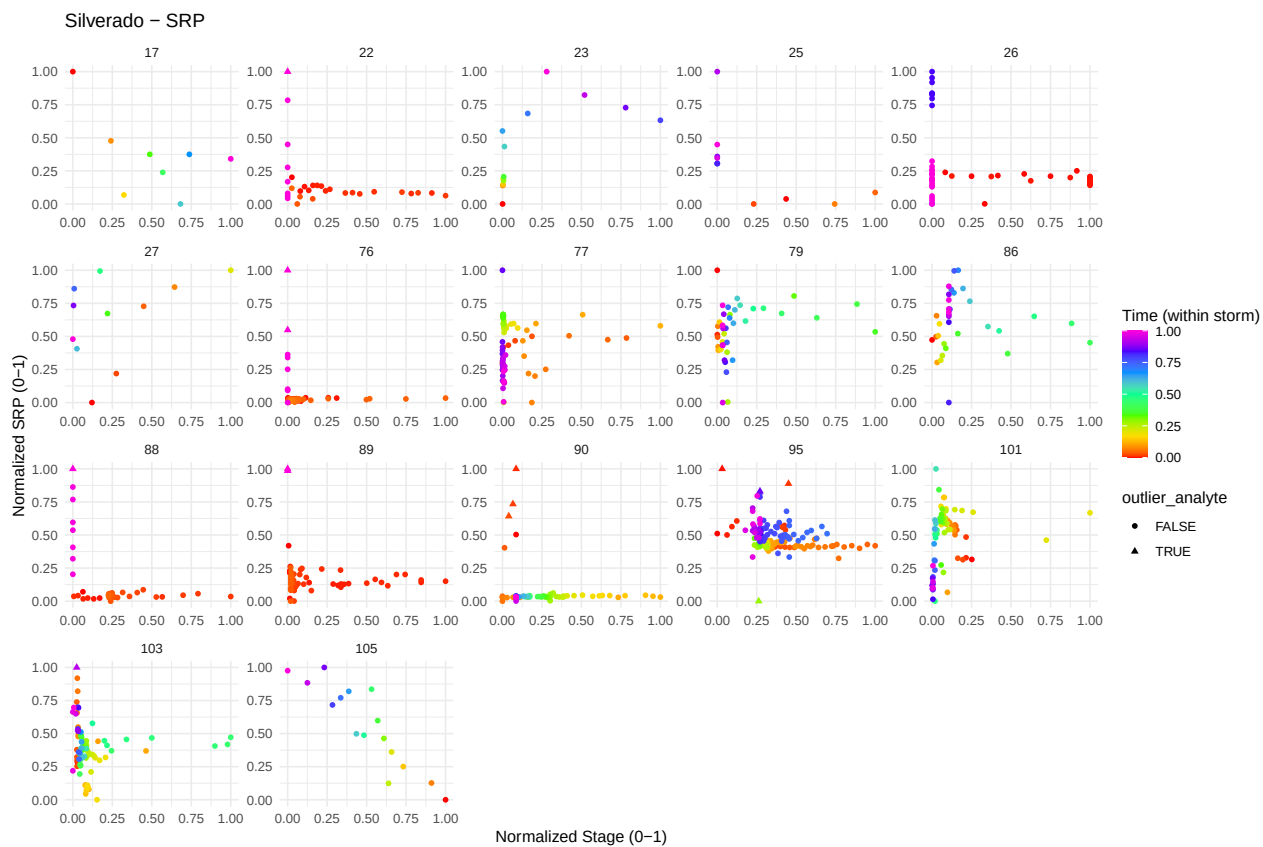


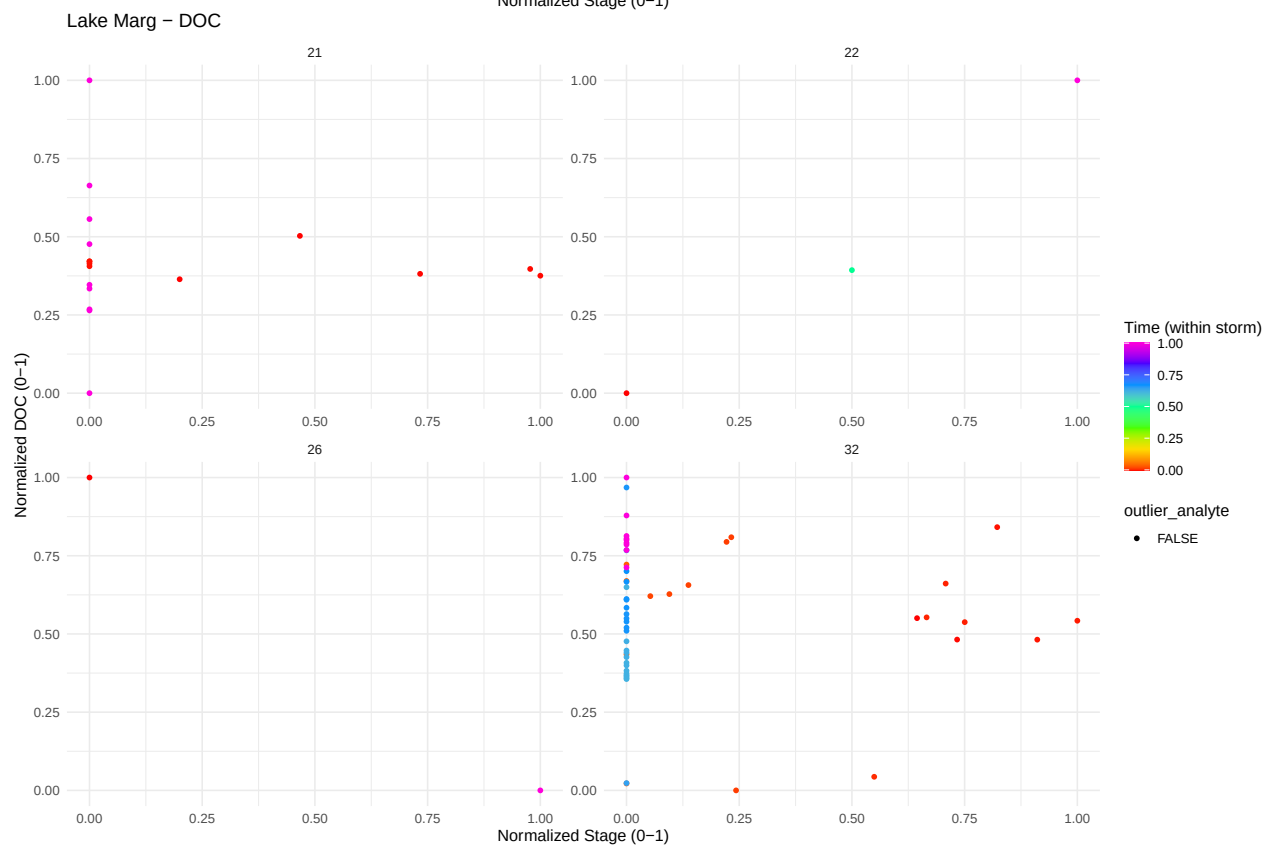
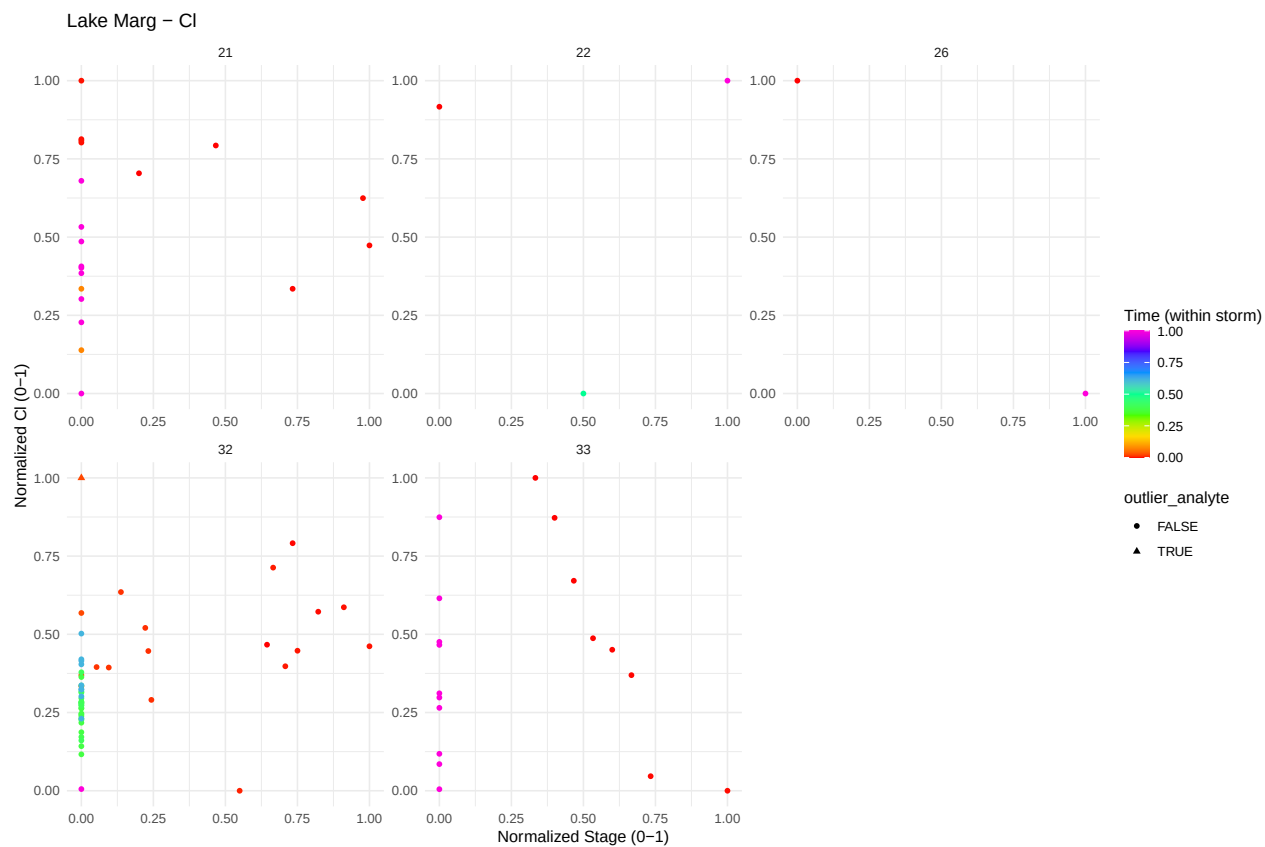


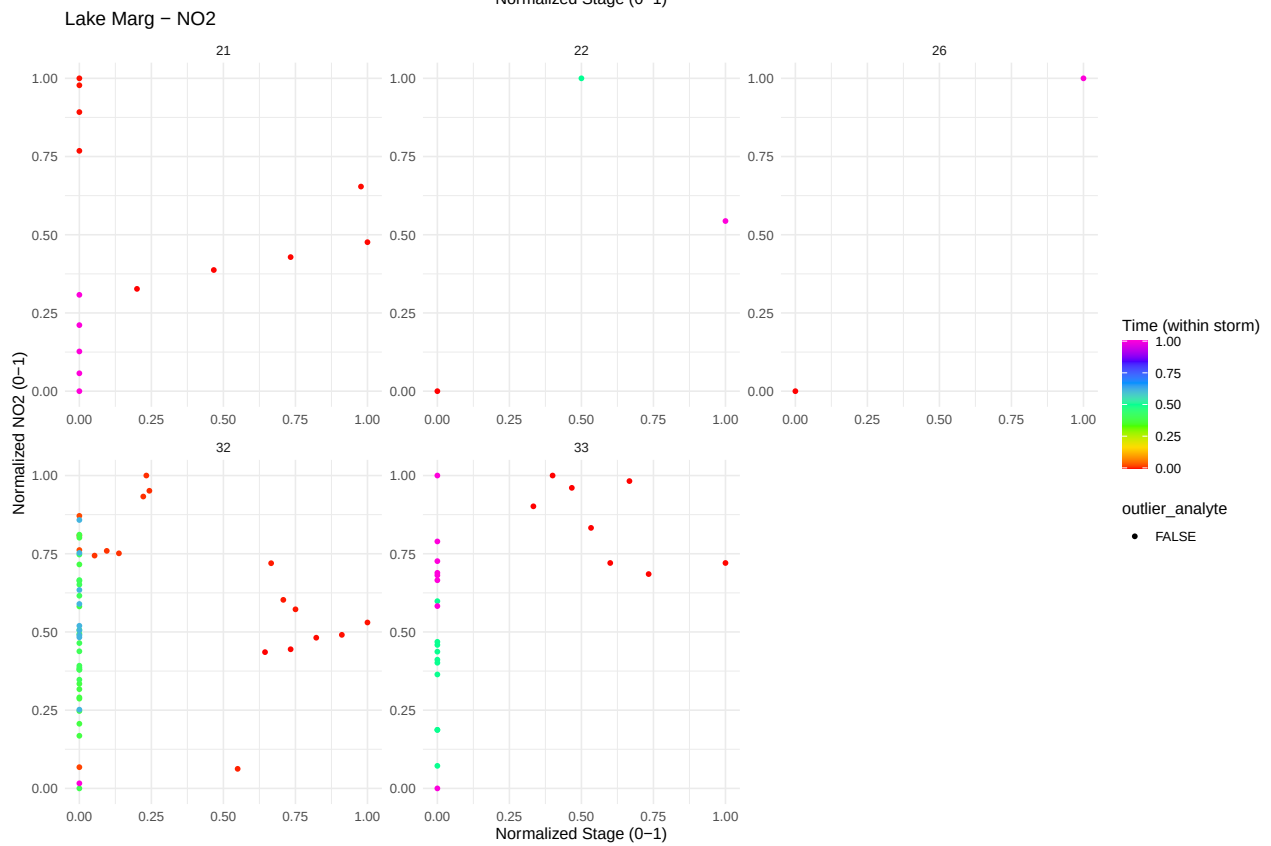
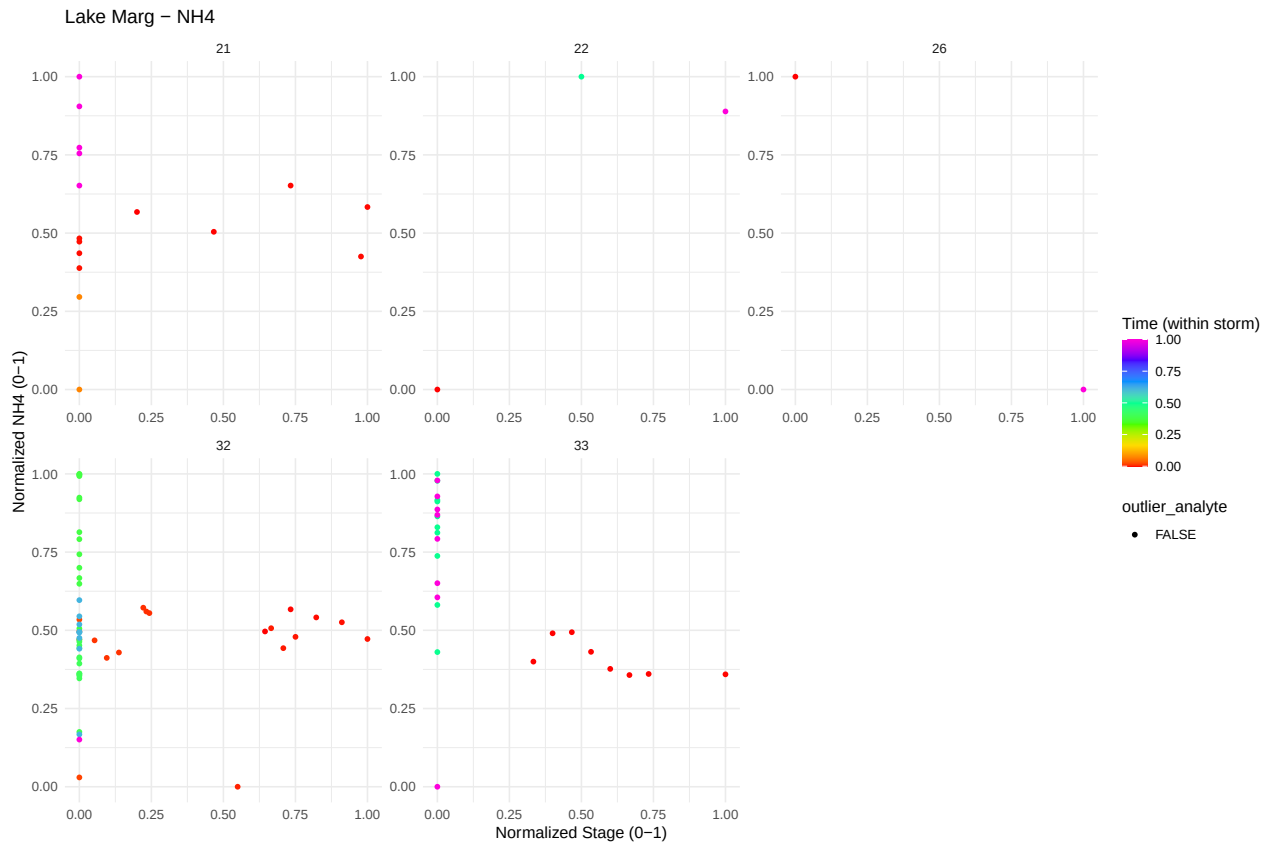


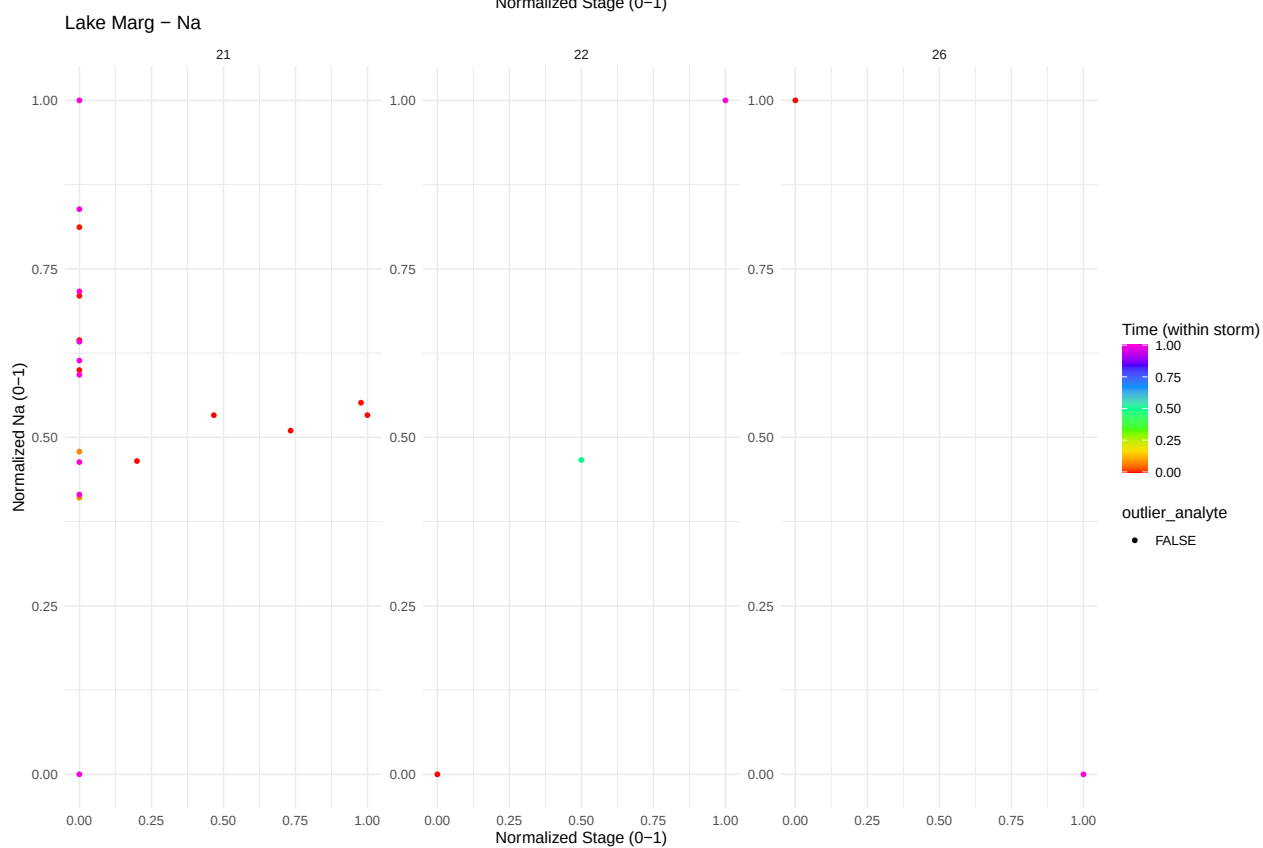
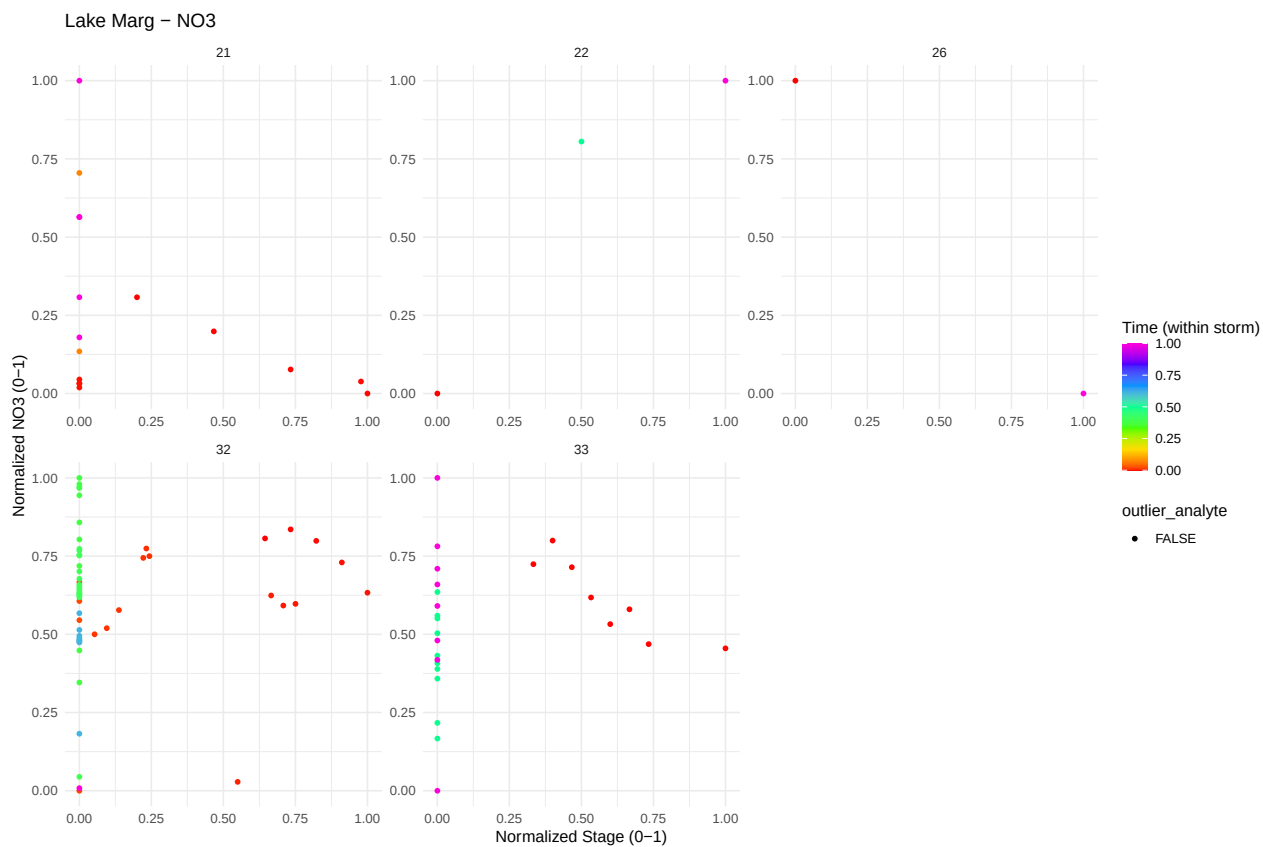


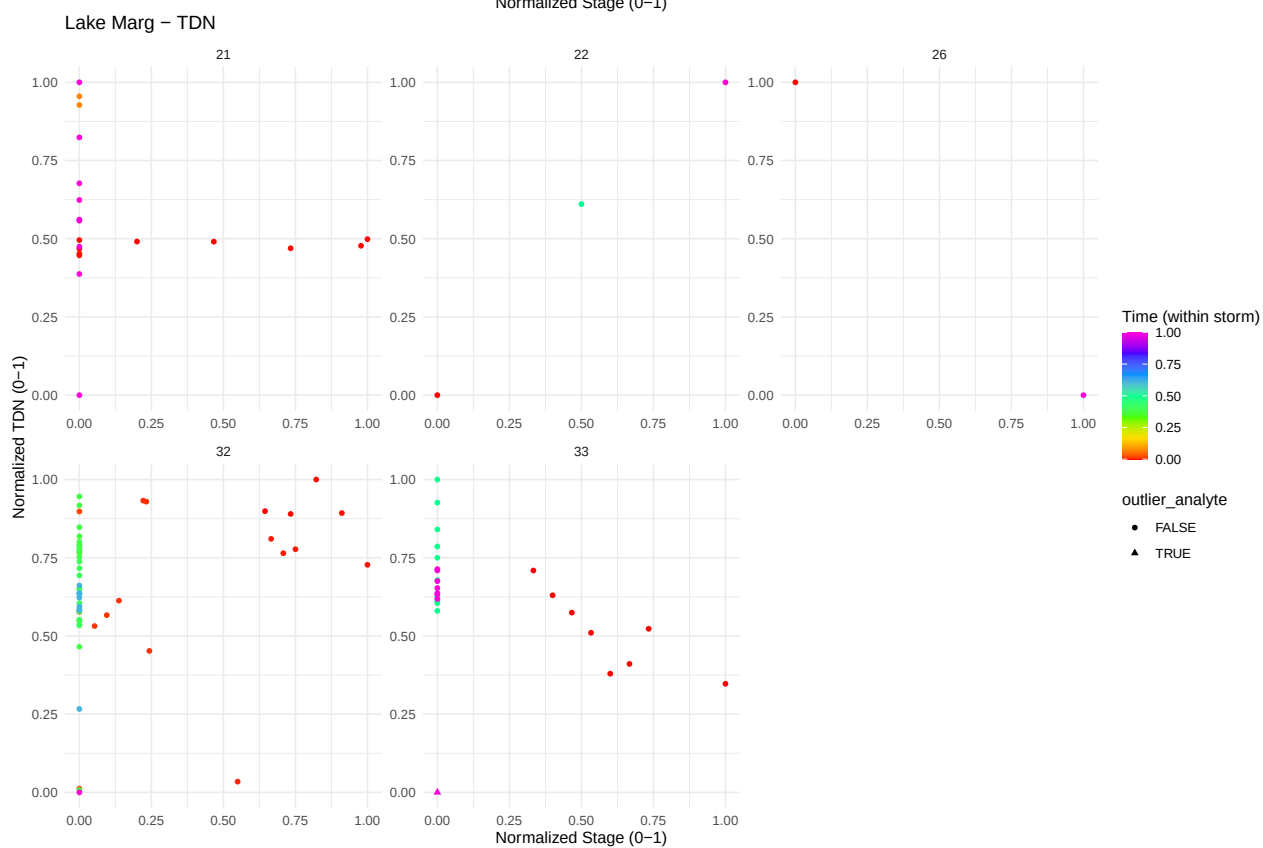
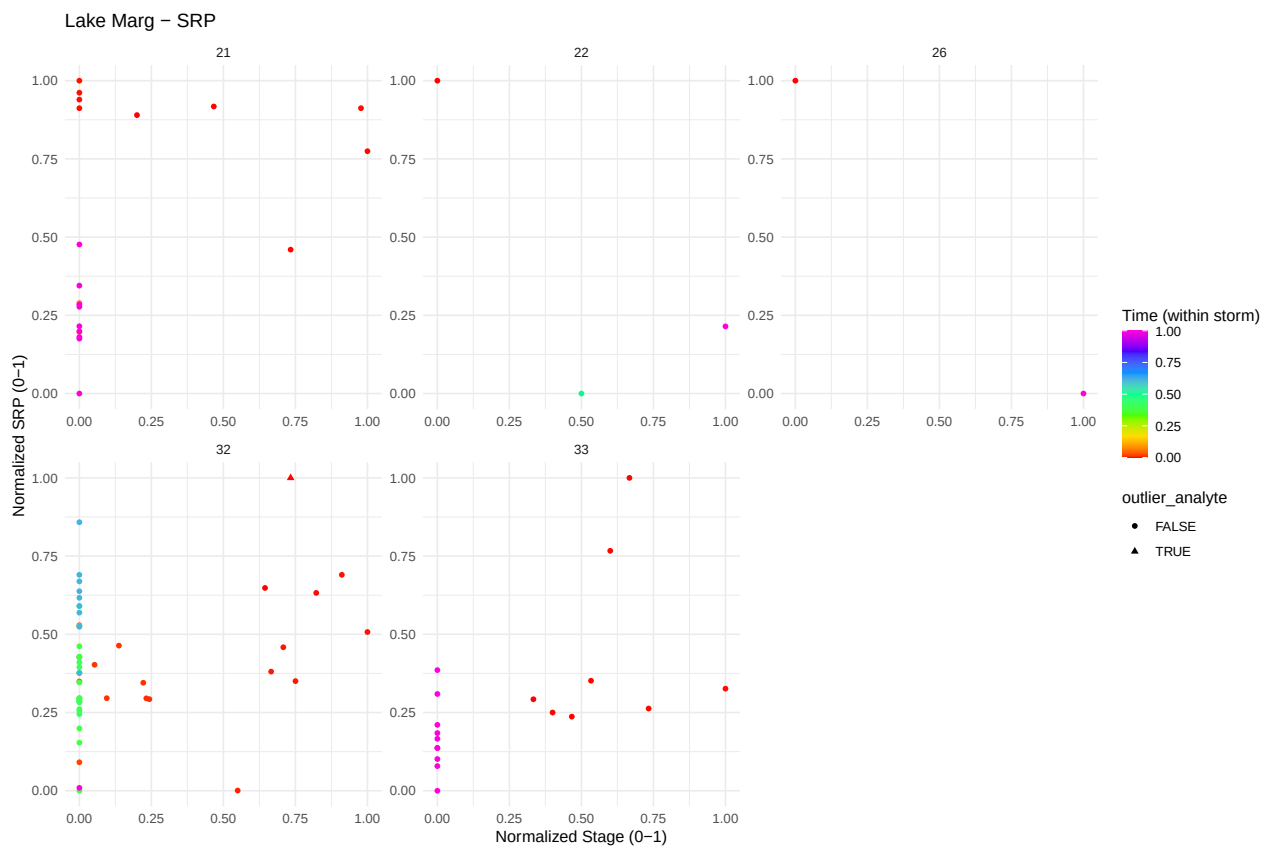


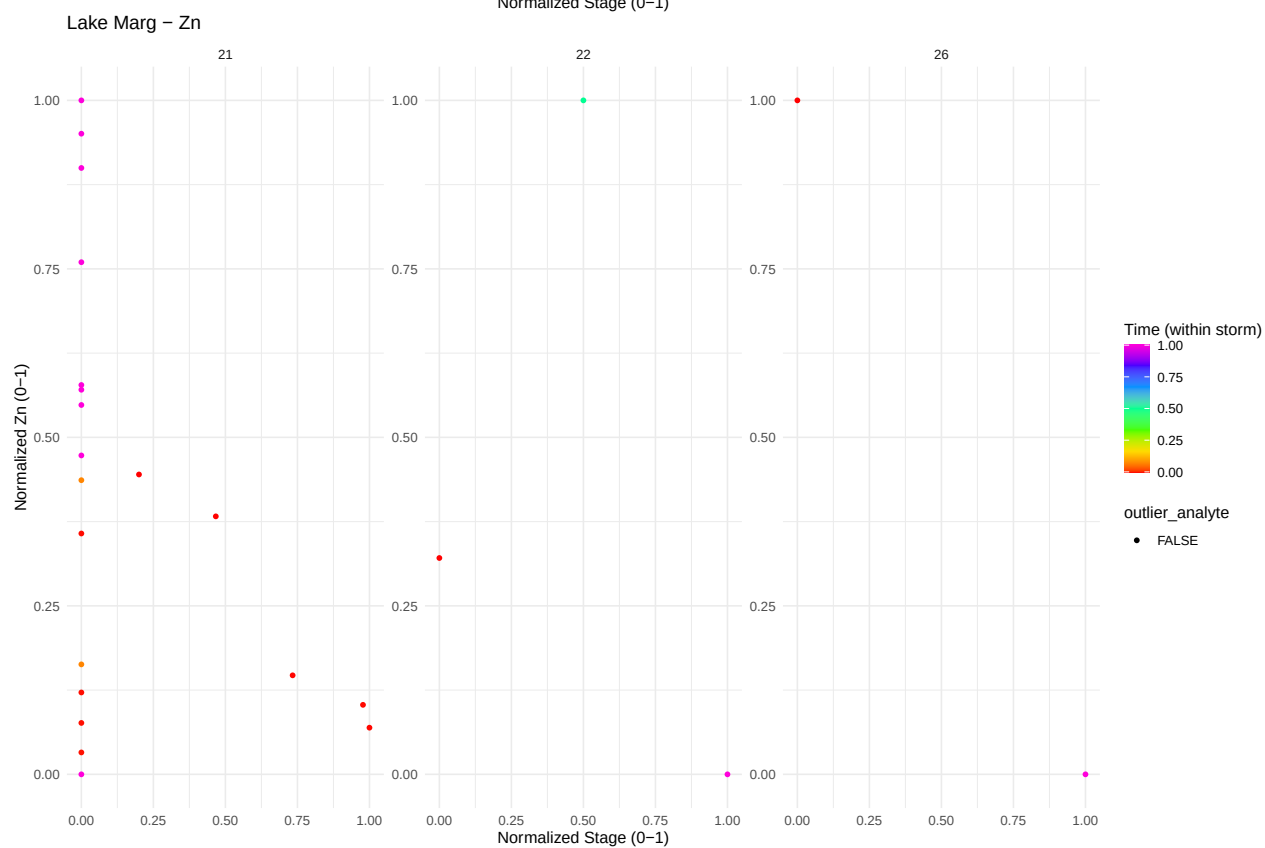
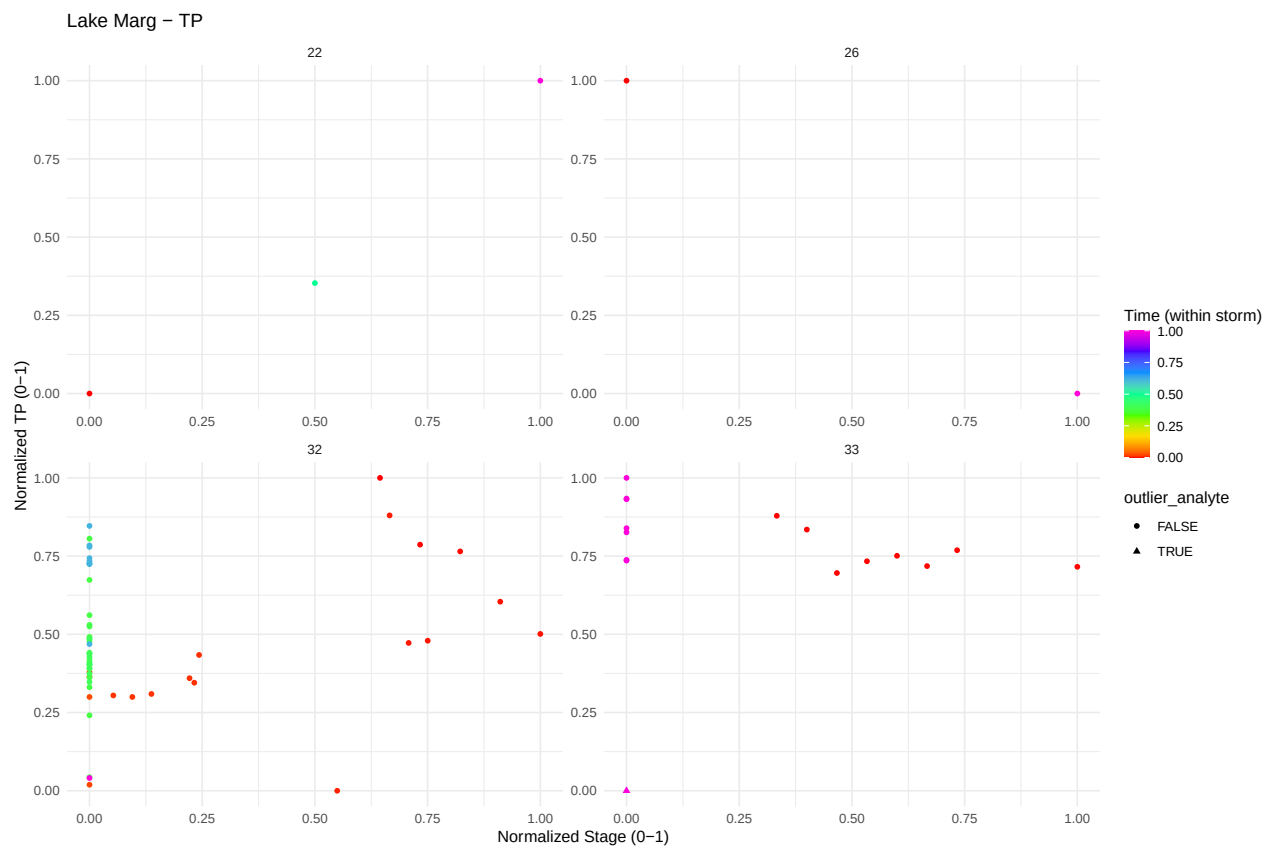












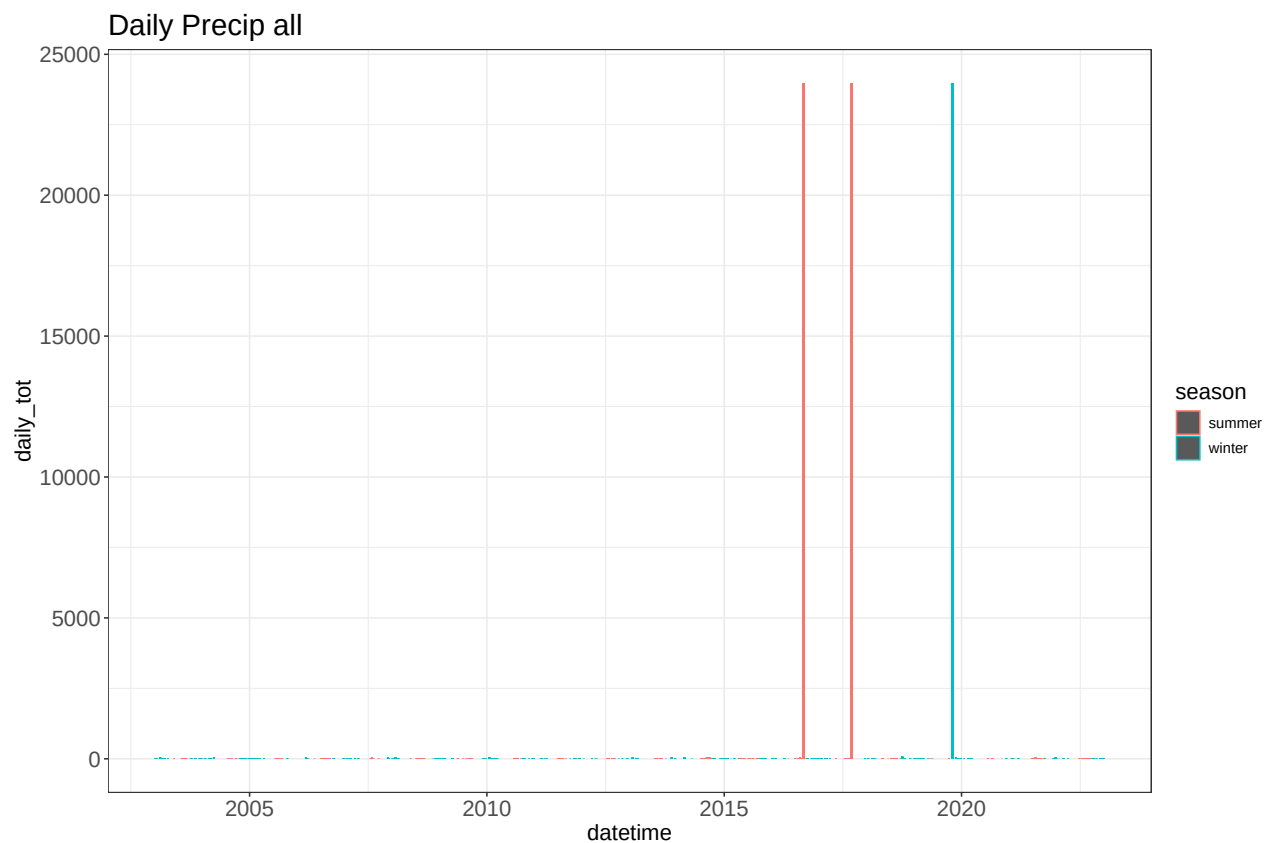
Precipitation

Precipitation data was collected from <https://azmet.arizona.edu/azmet/27.htm> - precipitation gage at the Desert Ridge station

Time series of precip

Daily Precip (excluding large outliers)

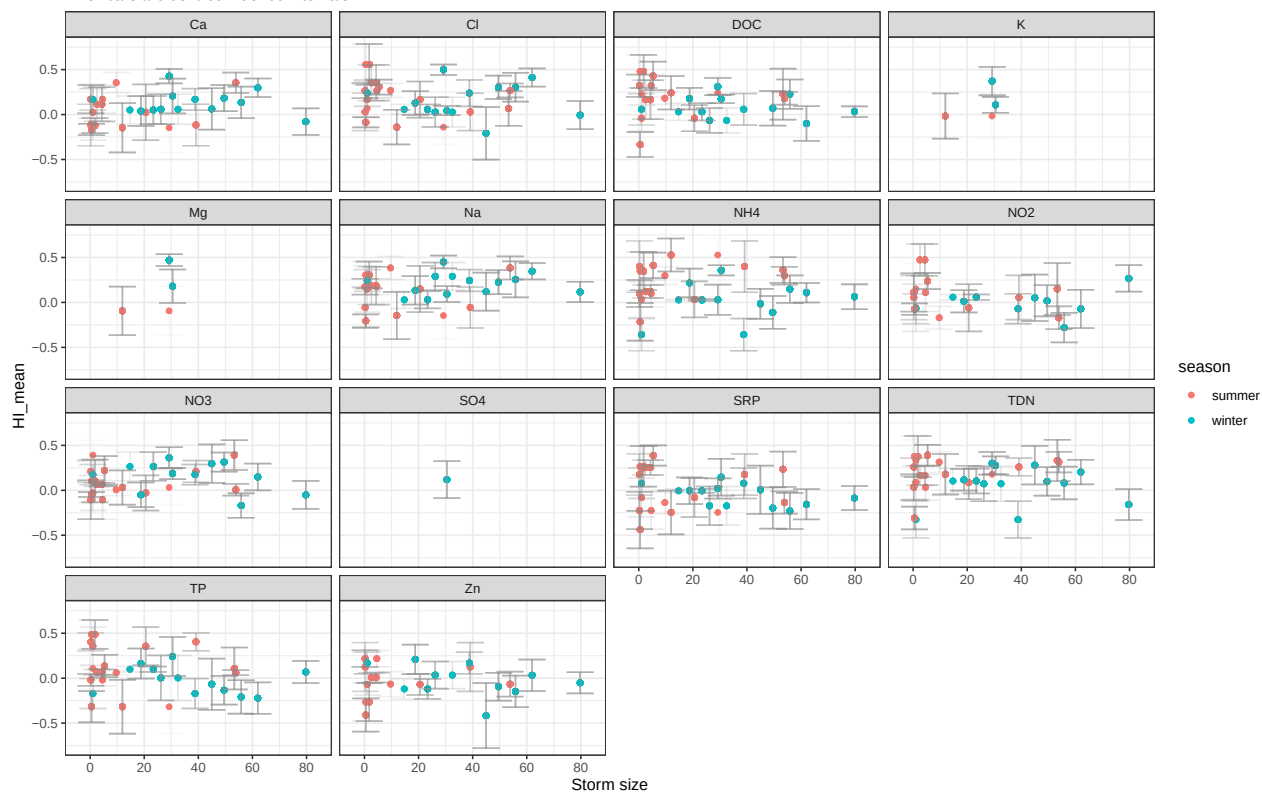




Storm size = cumulative precipitation (mm) per storm

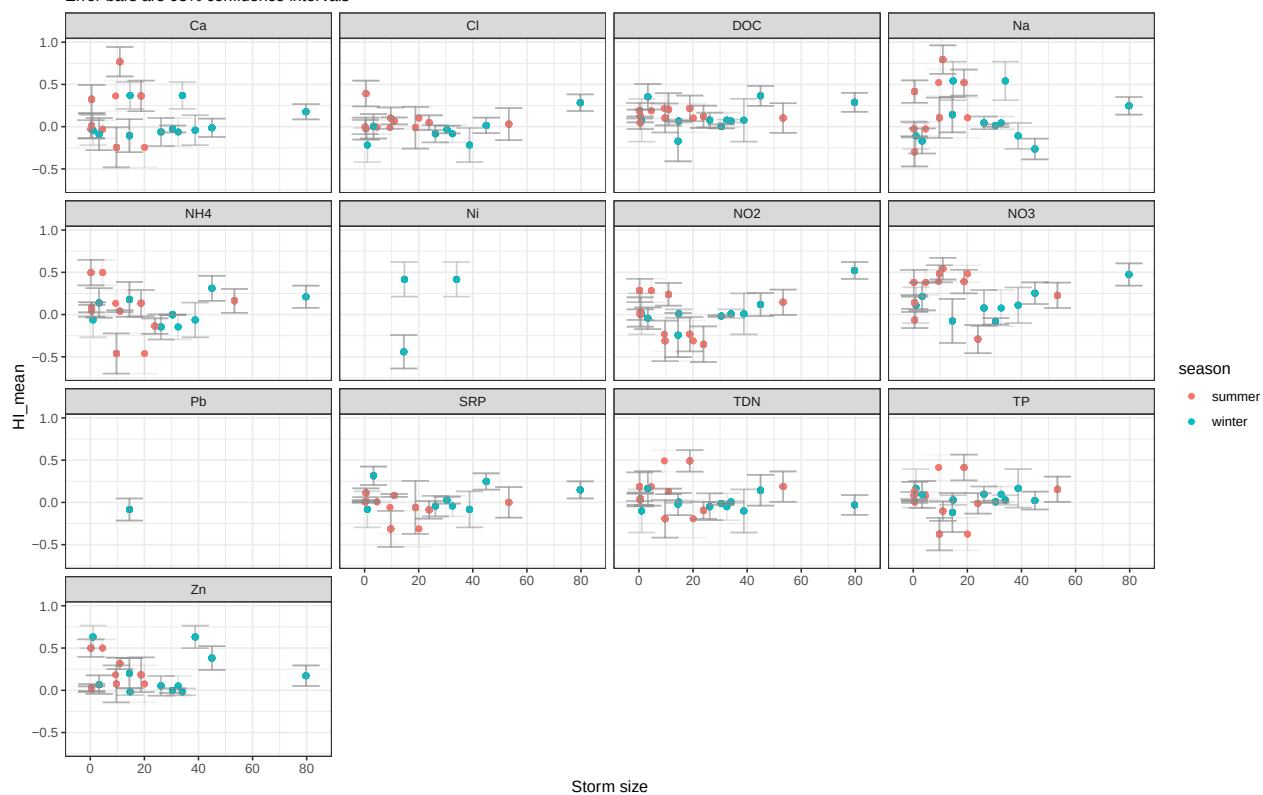
Curry HI mean vs cumulative precip per storm

Error bars are 95% confidence intervals



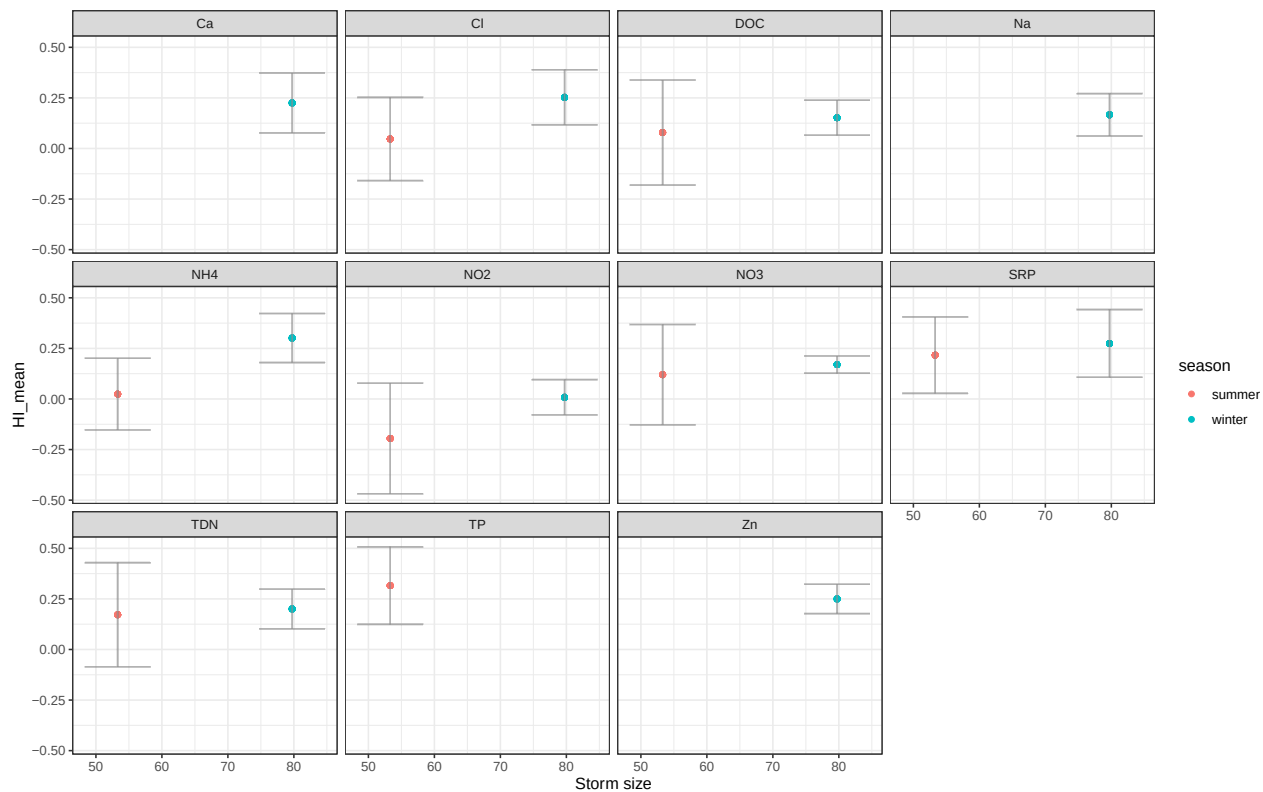
Silverado HI mean vs cumulative precip per storm

Error bars are 95% confidence intervals



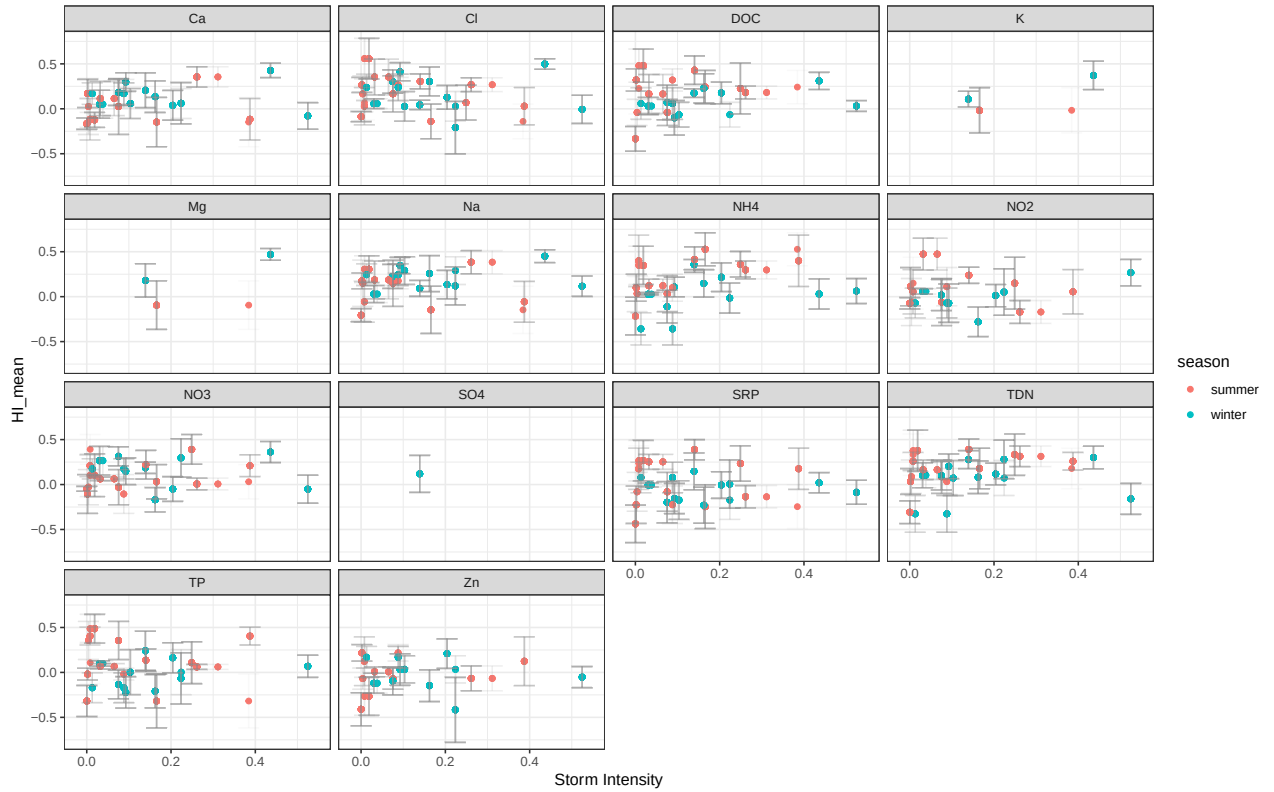
Lake Margarita HI mean vs cumulative precip per storm

Error bars are 95% confidence intervals

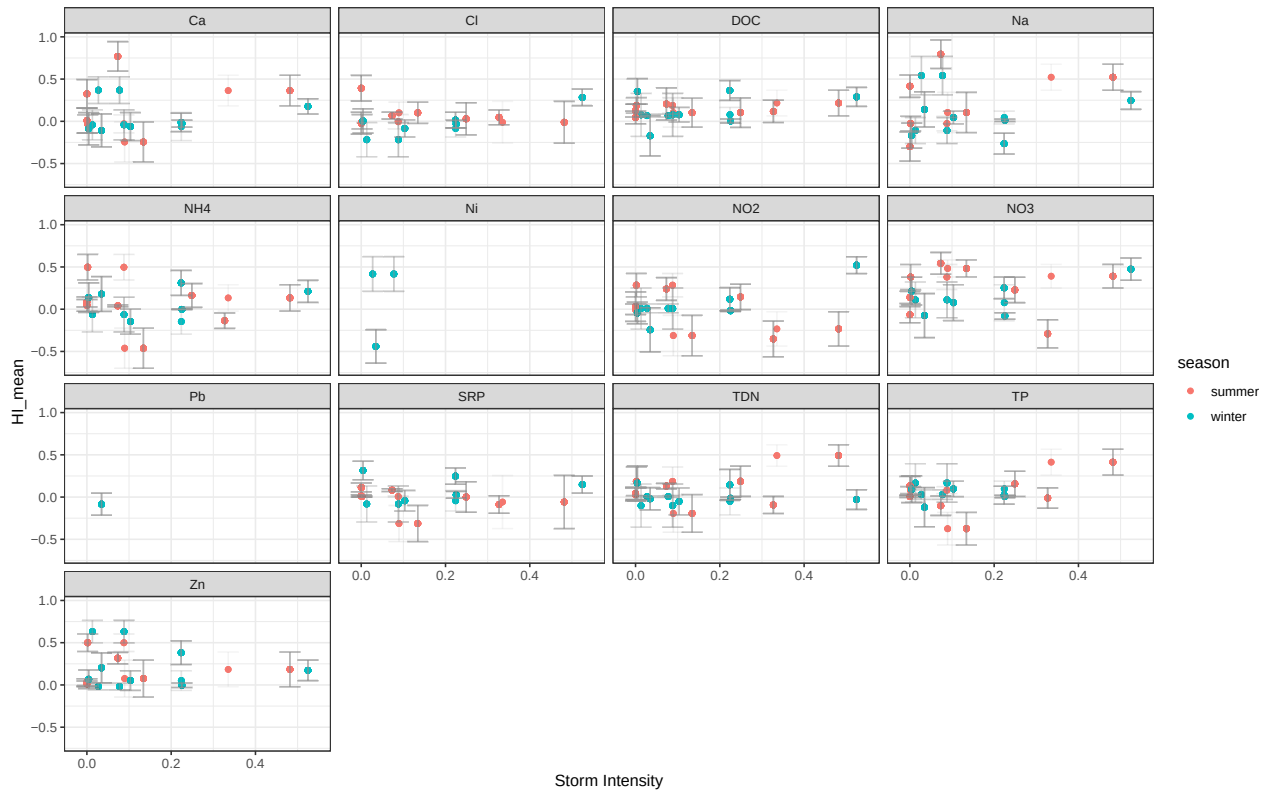


Storm intensity = cumulative precipitation/storm duration

Curry HI mean vs storm intensity
Error bars are 95% confidence intervals



Silverado HI mean vs storm intensity
Error bars are 95% confidence intervals



Lake Margarita HI mean vs storm intensity

Error bars are 95% confidence intervals

